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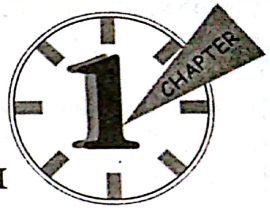
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POWER SYSTEM MODELLING



1.1 REPRESENTATION OF POWER SYSTEM

A typical power system consists of a 3-phase grid to which all generating stations feeds energy and from which all substations taps energy. (A grid is either a 3-phase single circuit or 3-phase two circuit transmission line, running throughout the length and breadth of a country or a state). From the substations electrical energy is transmitted to distribution transformers and from the distribution transformer, the energy is fed to various loads.

The components of power system are Generating stations (Alternators), Power transformers, Transmission lines, Substations (Substation transformers), Distribution transformers and Loads. The various types of loads are Synchronous motors, Induction motors, Heating coils, Lights, etc.,

The various components of power system and their interconnections are usually represented by single line diagram. In a single line diagram the components are represented by standard symbols and their interconnections are shown by single line, eventhough they are three phase circuits.

Single line diagram (One-Line Diagram)

A balanced three-phase system is always analysed on per phase basis by considering one of the three phase lines and neutral. Hence it is enough if we show one phase and neutral in the diagrammatic representation of power system. The diagram is further simplified by omitting the neutral and so the resultant diagram will be a single line diagram.

In single line representation of power system, the components of the system are represented by standard symbols and the transmission lines are represented by straight lines. Hence a single line diagram is diagrammatic representation of power system in which the components are represented by their symbols and the interconnections between them are shown by straight lines. Besides the symbols, the ratings and the impedances of the components are also marked on the single line diagram.

The purpose of the one-line diagram is to supply in concise form the significant information about the system. The various symbols used in single line diagram are shown in table-1.1. A typical single line diagram is shown in fig 1.1.

Table-1.1 Symbols used in single line diagram

Machine or rotating armature	◯	Power circuit breaker (oil/gas filled)	
Two-winding power transformer		Air circuit breaker	
Three-winding power transformer		Three-phase, three-w delta connection	△
Fuse		Three-phase star, neut ungrounded	Y
Current transformer		Three-phase star, neut grounded	
Potential transformer	or	Ammeter and voltmeter	A V

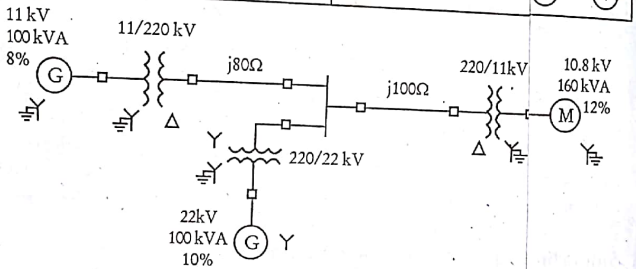


Fig 1.1 : A typical single line diagram

1.2 PER UNIT QUANTITIES

The electric power transmission lines are operated at very high voltage levels and transmits large amount of power. Hence the operating voltage of transmission line is expressed in kilovolt (kV) and power transmitted is expressed in kilowatt (kW) or megawatt (MW) and kilovolt ampere (kVA) or megavoltampere (MVA).

The various components of power system like alternators, motors, transformers, etc., have their voltage, power, current and impedance ratings in kV, kVA, kA and Ω respectively.

The components or various sections of power system may operate at different voltage and power levels. It will be convenient for analysis of power system if the voltage, power, current and impedance ratings of components of power system are expressed with reference to a common value called base value. Hence for analysis purpose a base value is chosen for voltage, power, current and impedance. Then all the voltage, power, current and impedance ratings of the components are expressed as a percent or perunit of the base value.

The per unit value of any quantity is defined as the ratio of the actual value of the quantity to the base value expressed as a decimal. The ratio in percent is 100 times the value in per unit. The base value is an arbitrary chosen value of the quantity.

$$\text{Per unit value} = \frac{\text{Actual value}}{\text{Base value}} \qquad \text{.....(1.1)}$$

$$\%\ \text{Per unit value} = \frac{\text{Actual value}}{\text{Base value}} \times 100 \qquad \text{.....(1.2)}$$

The power system requires the base values of four quantities and they are Voltage, Power, Current and Impedance. Selection of base values for any two of them determines the base values of the remaining two.

Single phase system

Let kVA_b = Base kVA
kV_b = Base voltage in kV
I_b = Base current in Amp
Z_b = Base impedance in Ω.

The following formulae relate the various quantities

$$\text{Base current, } I_b = \frac{kVA_b}{kV_b} \text{ in amps} \qquad \text{.....(1.3)}$$

$$\text{Base impedance, } Z_b = \frac{kV_b \times 1000}{I_b} \text{ in } \Omega \qquad \text{.....(1.4)}$$

On substituting for I_b from equation (1.3) in equ(1.4) we get,

$$\text{Base impedance, } Z_b = \frac{kV_b \times 1000}{\frac{kVA_b}{kV_b}} = \frac{(kV_b)^2}{kVA_b/1000} = \frac{kV_b^2}{MVA_b} \qquad \text{.....(1.5)}$$

$$\text{Per unit impedance} = \frac{\text{Actual impedance, } \Omega}{\text{Base impedance, } \Omega}$$

Three-phase system

The per-unit value of a line-to-neutral (V_{LN}) on the line-to-neutral voltage base ($V_{b,LN}$) is equal to the per-unit value of the line-to-line voltage (V_{LL}) at the same point on the line-to-line voltage base ($V_{b,LL}$) if the system is balanced,

$$\text{i.e., } \frac{V_{LN}}{V_{b,LN}} = \frac{V_{LL}}{V_{b,LL}} \qquad \text{.....(1.6)}$$

The per unit value of a 3-phase kVA on the 3-phase kVA base is identical to the per unit value of the kVA per phase on the kVA per phase base

$$\text{i.e., } \frac{\text{3-phase kVA}}{\text{3-phase base kVA}} = \frac{\text{kVA per phase}}{\text{Base kVA per phase}} \qquad \text{.....(1.7)}$$

Therefore in 3-phase systems the line value of voltages and 3-phase kVA are directly used for per unit calculations.

The base impedance and base current of 3-phase system can be computed directly from 3-phase value of base kVA and line value of base kV.

Let, kV_b = Line to line base kV

kVA_b = 3-phase base kVA

I_b = Line value of base current

Now, $kVA_b = \sqrt{3} \times kV_b \times I_b$

$$\left(\because \text{In 3-phase systems, } kVA = \sqrt{3}\, V_L I_L \times 10^{-3} = \sqrt{3} \times (kV_L)\, I_L\right) \qquad \text{.....(1.8)}$$

From equ(1.8) we get,

$$\text{Base current, } I_b = \frac{kVA_b}{\sqrt{3} \times kV_b} \qquad \text{.....(1.9)}$$

In a balanced power system the phase voltage is $1/\sqrt{3}$ times, the line voltage. Hence the base impedance per phase is given by

$$\left.\begin{matrix}\text{Base impedance}\\ \text{per phase}\end{matrix}\right\} Z_b = \frac{(kV_b/\sqrt{3}) \times 1000}{I_b} = \frac{kV_b \times 1000}{\sqrt{3}\, I_b} \qquad \text{.....(1.10)}$$

On substituting for I_b from equ(1.9) in equ(1.10) we get,

$$\left.\begin{matrix}\text{Base impedance}\\ \text{per phase}\end{matrix}\right\} Z_b = \frac{kV_b \times 1000}{\sqrt{3} \times \frac{kVA_b}{\sqrt{3} \times kV_b}} = \frac{(kV_b)^2}{\frac{kVA_b}{1000}} = \frac{(kV_b)^2}{MVA_b} \qquad \text{.....(1.11)}$$

Here, the equ(1.5) and (1.11) looks similar, but in 3-phase system, the kV_b is a line value and MVA_b is a 3-phase MVA.

Note : The impedance is always expressed as phase value.

Changing the base of per-unit quantities

The impedance of a device or component is usually specified in per unit on the base of name plate rating. When a system is formed by interconnecting various devices, it will be convenient for analysis if the impedances are converted to common base. Since all impedance in any one part of a system must be expressed on the common impedance base. It is necessary to have means of converting per-unit impedances from one base to another.

Let, Z = Actual impedance, Ω

Z_b = Base impedance, Ω

$$\text{Per unit impedance of a circuit element} = \frac{Z}{Z_b} = \frac{Z}{\frac{(kV_b)^2}{MVA_b}} = \frac{Z \times MVA_b}{(kV_b)^2} \qquad \text{.....(1.12)}$$

The equ(1.12) show that per unit impedance is directly proportional to base megavolt amperes and inversely proportional to the square of the base voltage. Using equ(1.12) we can derive an expression to convert the p.u. impedance expressed in one base value (old base) to another base (new base).

Let $kV_{b,old}$ and $MVA_{b,old}$ represents old base values and $kV_{b,new}$ and $MVA_{b,new}$ represents new base value.

Let, $Z_{pu,old}$ = p.u. impedance of a circuit element calculated on old base.

$Z_{pu,new}$ = p.u. impedance of a circuit element calculated on new base.

If old base values are used to compute the p.u. impedance of a circuit element with impedance Z, then equ(1.12) can be written as,

$$Z_{pu,old} = \frac{Z \times MVA_{b,old}}{(kV_{b,old})^2} \qquad \text{.....(1.13)}$$

$$Z = Z_{pu,old} \frac{(kV_{b,old})^2}{MVA_{b,old}} \qquad \text{.....(1.14)}$$

If the new base values are used to compute the p.u. impedance of a circuit element with impedance Z, then equ(1.12) can be written as

$$Z_{pu,new} = \frac{Z \times MVA_{b,new}}{(kV_{b,new})^2} \qquad \text{.....(1.15)}$$

On substituting for Z from equ(1.14) in equ(1.15) we get,

$$Z_{pu,\,new} = Z_{pu,\,old}\frac{(kV_{b,\,old})^2}{MVA_{b,\,old}} \times \frac{MVA_{b,\,new}}{(kV_{b,\,new})^2}$$

$$= Z_{pu,\,old} \times \left(\frac{kV_{b,\,old}}{kV_{b,\,new}}\right)^2 \times \left(\frac{MVA_{b,\,new}}{MVA_{b,\,old}}\right) \quad \text{.....(1.16)}$$

The equ(1.16) can be used to convert the p.u. impedance expressed on one base value to another base.

Advantages of per-unit computations

1. Manufacturers usually specify the impedance of a device or machine in percent or per unit on the base of the name plate rating.
2. The per-unit impedances of machines of the same type and widely different rating usually lie within a narrow range, although the ohmic values differ widely for machines of different ratings.
3. The per-unit impedance of circuit element connected by transformers expressed on a proper base will be same if it is referred to either side of a transformer.
4. The way in which the transformers are connected in 3-phase circuits (Y or Δ) does not affect the per-unit impedances of the equivalent circuit, although the transformer connection does determine the relation between the voltage bases on the two sides of the transformer.

1.3 EQUIVALENT CIRCUITS OF COMPONENTS OF POWER SYSTEM

The equivalent circuit of a power system is needed to perform analysis like load flow analysis, fault level calculations, etc. It can be obtained from the equivalent circuit of the components of the power system. The various components of power system are generator (alternator), transformer, transmission line, induction motor, synchronous motor, resistive and reactive loads. The equivalent circuits of various electrical machines developed in electrical machine theory can be used in power system modelling with or without approximations.

Equivalent circuit of generator

The equivalent circuit of a 3-phase generator (Alternator) is shown in fig 1.2. It consists of a source representing induced emf per phase, a series reactance representing the armature reactance and leakage reactance and a series resistance representing the armature winding resistance.

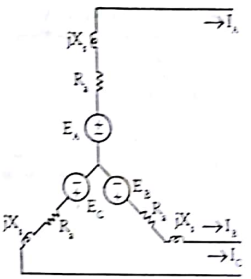


Fig a : 3-Phase equivalent circuit

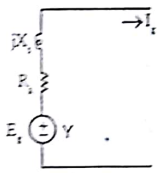


Fig b : Single phase equivalent circuit

Fig 1.2 : Equivalent circuit of generator (3-phase alternator)

Equivalent circuit of synchronous motor

The equivalent circuit of synchronous motor is shown in fig 1.3. The synchronous motor is similar to a generator in construction, but it performs the reverse action of the generator. (A generator converts mechanical energy to electrical energy, but the motor converts electrical energy to mechanical energy). Therefore the direction of current in motor is opposite to that of generator.

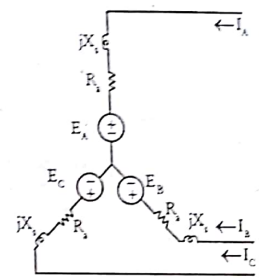


Fig a : 3-Phase equivalent circuit

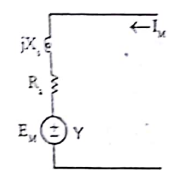


Fig b : Single phase equivalent circuit

Fig 1.3 : Equivalent circuit of synchronous motor

Equivalent circuit of transformer

The equivalent circuit of a single phase, two winding transformer referred to primary is shown in fig 1.4. It consists of shunt branches to represent magnetising current and core loss, series resistance representing winding resistance referred to primary and the series reactance representing leakage reactance referred to primary.

$$K = \frac{E_2}{E_1} = \frac{N_2}{N_1} \approx \frac{V_2}{V_1} = \frac{I_1}{I_2}$$

$$R_{01} = R_1 + R_2' = R_1 + R_2 / K^2$$

$$X_{01} = X_1 + X_2' = X_1 + X_2 / K^2$$

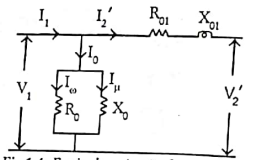


Fig 1.4 : Equivalent circuit of a transformer

The three phase transformer is represented by its single phase equivalent and the equivalent circuit is similar to that of fig 1.4. In three phase transformers, the transformation ratio, K is taken as the ratio of line voltages. This will facilitate the direct conversion of star side impedances to delta side and vice versa.

Equivalent circuit of induction motor

The single phase equivalent circuit of induction motor referred to stator is shown in fig 1.5. It is similar to equivalent circuit of transformer (The induction motor is also called rotating transformer).

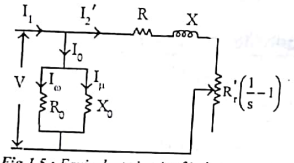


Fig 1.5 : Equivalent circuit of induction motor

s = Slip

$R_r'\left(\frac{1}{s} - 1\right)$ = Resistance representing load.

R = $R_s + R_r'$ = Equivalent resistance referred to stator

X = $X_s + X_r'$ = Equivalent reactance referred to stator

R_s, X_s = Resistance and reactance of stator

R_r, X_r = Resistance and reactance of rotor.

Equivalent circuit of transmission line

The transmission line can be represented by its resistance, inductance and capacitance. The single phase equivalent π-type and T-type model of the transmission line is shown in fig 1.6. The elements R, X_L and X_c are resistance, inductive reactance and capacitive reactance per phase respectively.

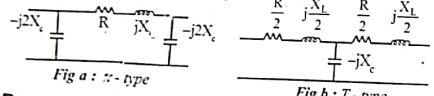


Fig a : π - type *Fig b : T - type*

Fig 1.6 : Equivalent circuit of transmission line

Representation of resistive and reactive loads

The resistive and reactive loads can be represented in the equivalent circuit by any one of the following

1. Constant power representation
2. Constant current representation ; 3. Constant impedance representation

In constant power representation, the load active power (MW) and reactive power (MVAR) are considered to be constant. This method of representation will be useful in load flow studies.

In constant current representation the magnitude of the load current is considered as constant. The constant current of the load can be calculated from the specified voltage, active and reactive powers of the load, as shown below.

Single phase load

Let P = Active power
and Q = Reactive power

The complex power, $S = VI^* = P + jQ$

$$\therefore (VI^*)^* = (P + jQ)^*$$

$$V^*I = P - jQ$$

$$\therefore I = \frac{P - jQ}{V^*} \quad \text{.....(1.17)}$$

Let, $V = |V| \angle\delta$; $V^* = |V| \angle -\delta$ .....(1.18)

and $P - jQ = \sqrt{P^2 + Q^2} \angle -\tan^{-1}\frac{Q}{P} = \sqrt{P^2 + Q^2} \angle -\theta$

where $\theta = \tan^{-1}\frac{Q}{P}$ .....(1.19)

From equations (1.17), (1.18) and (1.19) we can write

$$I = \frac{\sqrt{P^2 + Q^2} \angle -\theta}{|V| \angle -\delta} = \frac{\sqrt{P^2 + Q^2}}{|V|} \angle \delta - \theta = |I| \angle \delta - \theta \quad \text{.....(1.20)}$$

where $|I| = \frac{\sqrt{P^2 + Q^2}}{|V|}$ .....(1.21)

In constant impedance representation the load is represented by its impedance or admittance. The impedance of the load can be calculated from the specified voltage, active and reactive powers of the load as shown below.

Load Impedance, $Z = \frac{V}{I}$ .....(1.22)

On substituting for I from equ(1.17) in equ(1.22) we get,

$$\therefore \text{Load impedance, } Z = \frac{V}{(P - jQ)/V^*} = \frac{V V^*}{P - jQ} = \frac{|V|^2}{P - jQ} \quad \text{.....(1.23)}$$

$$\text{Load admittance, } Y = \frac{1}{Z} = \frac{P - jQ}{|V|^2} \quad \text{.....(1.24)}$$

Three phase load

Balanced star connected load

Let P = Three phase active power of star connected load in watts.
Q = Three phase reactive power of star connected load in VARs.
V, V_L = Phase & line voltage of load respectively.
I, I_L = Phase & line current of load respectively.

Three phase complex power, $S = 3VI^* = P + jQ$

$$\therefore (3VI^*)^* = (P + jQ)^*$$

$$3V^*I = P - jQ$$

$$\therefore I = \frac{P - jQ}{3V^*} \quad \text{.....(1.25)}$$

Let $V = |V|\angle\delta$; $\therefore V^* = |V|\angle -\delta$

In star connected load, $|V| = \frac{|V_L|}{\sqrt{3}}$ and $I = I_L$

$$\therefore V^* = \frac{|V_L|}{\sqrt{3}}\angle -\delta \quad \text{.....(1.26)}$$

$$\text{Let } P - jQ = \sqrt{P^2 + Q^2}\angle -\tan^{-1}\frac{Q}{P} = \sqrt{P^2 + Q^2}\angle -\theta \quad \text{.....(1.27)}$$

where $\theta = \tan^{-1}\frac{Q}{P}$

Using equations (1.26) and (1.27), the equation (1.25) can be written as,

$$I = \frac{P - jQ}{3V^*} = \frac{\sqrt{P^2 + Q^2}\angle -\theta}{3\frac{|V_L|}{\sqrt{3}}\angle -\delta} = \frac{\sqrt{P^2 + Q^2}}{\sqrt{3}|V_L|}\angle \delta - \theta$$

$$\therefore I = I_L = \frac{\sqrt{P^2 + Q^2}}{\sqrt{3}|V_L|}\angle \delta - \theta \quad \text{.....(1.28)}$$

$$\therefore |I| = |I_L| = \frac{\sqrt{P^2 + Q^2}}{\sqrt{3}|V_L|} \quad \text{.....(1.29)}$$

Load impedance per phase, $Z = \frac{V}{I}$ .....(1.30)

On substituting for I from equ(1.25) in equ(1.30) we get,

$$Z = \frac{V}{\frac{(P - jQ)}{3V^*}} = \frac{3VV^*}{P - jQ} = \frac{3|V|^2}{P - jQ}$$

$$= \frac{3\left(|V_L|/\sqrt{3}\right)^2}{P - jQ} = \frac{|V_L|^2}{P - jQ}$$

$$\therefore \text{Load impedance per phase, } Z = \frac{|V_L|^2}{P - jQ} \quad \text{.....(1.31)}$$

$$\therefore \text{Load admittance per phase, } Y = \frac{1}{Z} = \frac{P - jQ}{|V_L|^2} \quad \text{.....(1.32)}$$

Balanced delta connected Load

Let P = Three phase active power of delta connected load in watts
Q = Three phase reactive power of delta connected load in VARs.
V, V_L = Phase & line voltage of load respectively.
I, I_L = Phase & line current of load respectively.

Three phase complex power, $S = 3VI^* = P + jQ$

$$\therefore (3VI^*)^* = (P + jQ)^*$$

$$3V^*I = P - jQ$$

$$\therefore I = \frac{P - jQ}{3V^*} \quad \text{.....(1.33)}$$

Let $V = |V|\angle\delta$; $\therefore V^* = |V|\angle -\delta$

In delta connected load, $V = V_L$ and $|I| = |I_L|/\sqrt{3}$ .....(1.34)

$$\text{Let, } P - jQ = \sqrt{P^2 + Q^2}\angle -\tan^{-1}\frac{Q}{P} = \sqrt{P^2 + Q^2}\angle -\theta \quad \text{.....(1.35)}$$

where $\theta = \tan^{-1}\frac{Q}{P}$

Using equations (1.34) and (1.35), the equation (1.33) can be written as,

$$I = \frac{\sqrt{P^2 + Q^2}\angle -\theta}{3|V|\angle -\delta} = \frac{\sqrt{P^2 + Q^2}}{3|V|}\angle(\delta - \theta) = \frac{\sqrt{P^2 + Q^2}}{3|V_L|}\angle(\delta - \theta)$$

$$= |I|\angle(\delta - \theta)$$

where, $|I| = \frac{\sqrt{P^2 + Q^2}}{3|V_L|}$

$$\therefore |I_L| = \sqrt{3}\,|I| = \sqrt{3}\,\frac{\sqrt{P^2+Q^2}}{3|V_L|} = \frac{\sqrt{P^2+Q^2}}{\sqrt{3}|V_L|} \quad \text{.....(1.36)}$$

Load impedance per phase, $Z = \frac{V}{I}$ .....(1.37)

On substituting for I from equ(1.33) in equ(1.37) we get,

$$Z = \frac{V}{\frac{(P-jQ)}{3V^*}} = \frac{3VV^*}{P-jQ} = \frac{3|V|^2}{P-jQ} = \frac{3|V_L|^2}{P-jQ}$$

Load impedance per phase, $Z = \frac{3|V_L|^2}{P-jQ}$ .....(1.38)

Load admittance per phase, $Y = \frac{1}{Z} = \frac{P-jQ}{3|V_L|^2}$ .....(1.39)

Three winding transformer

In addition to the primary and secondary winding, the transformer may be constructed with a third winding called tertiary winding. In three winding transformers, the two windings are connecting in star and one winding in delta or two windings in delta and one winding in star. The purpose of providing the tertiary winding are the following.

1. To get supply voltage for the substation auxiliary devices. [The auxiliary devices may work at a voltage different from those of the primary and secondary windings].
2. Static capacitors or synchronous condensers may be connected to the tertiary winding for reactive power injection into the system for voltage control.
3. A delta connected tertiary reduces the impedance offered to the zero sequence currents thereby allowing a large earth-fault current to flow for proper protection of protective equipments. Further it limits voltage imbalance when the load is unbalanced. It also permits the third harmonic current to flow thereby reducing third harmonic voltages. For these reason the tertiary winding is also called stabilization winding.
4. Three windings may be used for interconnecting three transmission line at different voltages.

Tertiary can serve the purpose of measuring voltage of an HV testing nsformer.

transformer, the three windings may have different kVA rating. ry and secondary of two winding transformers have the same kVA rating. nce of each winding of a three winding transformer may be given in p.u.

calculated by using their own winding rating as bases. But while representing in reactance/ impedance diagram it is necessary to convert the p.u. reactances to a common base.

The equivalent circuit of a three winding transformer can be represented by the single phase equivalent circuit shown in fig 1.7.

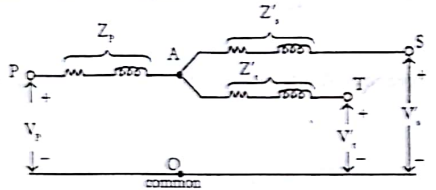


Fig 1.7 : Single phase equivalent circuit of three winding transformer

In this equivalent circuit the impedance per phase of the three windings (referred to one of winding, usually primary) are connected in star to represent the single phase equivalent circuit. For simplicity, the effect of exciting current is ignored in the equivalent circuit. The subscripts p, s and t indicate the primary, secondary and tertiary respectively. Three external circuits are connected between P & O, S & O and T & O, where the terminal O is the common terminal.

The impedances of the three windings are calculated using a common base kVA or MVA. The base voltage is the voltage rating of respective windings. The three impedances can be measured by the standard short-circuit tests.

Z_p = Impedance of primary winding

Z'_s = Impedance of secondary winding referred to primary

Z'_t = Impedance of tertiary winding referred to primary

Z_{ps} = Leakage impedance measured in primary with secondary short-circuited and tertiary open.

Z_{pt} = Leakage impedance measured in primary with tertiary short-circuited and secondary open.

Z'_{st} = Leakage impedance measured in secondary with tertiary short-circuited and primary open and then referred to primary.

The leakage impedances measured by short circuit test are related to winding impedance as follows.

$$Z_{ps} = Z_p + Z'_s \quad \text{.....(1.40)}$$

$$Z_{pt} = Z_p + Z'_t \quad \text{.....(1.41)}$$

$$Z'_{st} = Z'_s + Z'_t \quad \text{.....(1.42)}$$

The equations (1.40) to (1.42) can be expressed in the form of matrix equation as shown in equ(1.43)

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} Z_p \\ Z_s' \\ Z_t' \end{bmatrix} = \begin{bmatrix} Z_{ps} \\ Z_{pt} \\ Z_{st}' \end{bmatrix} \quad \text{.....(1.43)}$$

The matrix equation (1.43) can be solved by cramer's rule as shown below.

$$\Delta = \begin{vmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{vmatrix} = 1 \times (-1) - 1(1) = -2$$

$$\Delta_1 = \begin{vmatrix} Z_{ps} & 1 & 0 \\ Z_{pt} & 0 & 1 \\ Z_{st}' & 1 & 1 \end{vmatrix} = Z_{ps}(-1) - 1(Z_{pt} - Z_{st}') = -Z_{ps} - Z_{pt} + Z_{st}'$$

$$\Delta_2 = \begin{vmatrix} 1 & Z_{ps} & 0 \\ 1 & Z_{pt} & 1 \\ 0 & Z_{st}' & 1 \end{vmatrix} = 1 \times (Z_{pt} - Z_{st}') - Z_{ps}(1) = -Z_{ps} - Z_{st}' + Z_{pt}$$

$$\Delta_3 = \begin{vmatrix} 1 & 1 & Z_{ps} \\ 1 & 0 & Z_{pt} \\ 0 & 1 & Z_{st}' \end{vmatrix} = 1 \times (-Z_{pt}) - 1(Z_{st}') + Z_{ps}(1) = -Z_{pt} - Z_{st}' + Z_{ps}$$

$$Z_p = \frac{\Delta_1}{\Delta} = \frac{1}{-2}\left[-Z_{ps} - Z_{pt} + Z_{st}'\right] = \frac{1}{2}\left[Z_{ps} + Z_{pt} - Z_{st}'\right] \quad \text{.....(1.44)}$$

$$Z_s' = \frac{\Delta_2}{\Delta} = \frac{1}{-2}\left[-Z_{ps} - Z_{st}' + Z_{pt}\right] = \frac{1}{2}\left[Z_{ps} + Z_{st}' - Z_{pt}\right] \quad \text{.....(1.45)}$$

$$Z_t' = \frac{\Delta_3}{\Delta} = \frac{1}{-2}\left[-Z_{pt} - Z_{st}' + Z_{ps}\right] = \frac{1}{2}\left[Z_{pt} + Z_{st}' - Z_{ps}\right] \quad \text{.....(1.46)}$$

The equations (1.44) to (1.46) can be used to calculate the impedances of the three windings (referred to primary) using the short circuit test data.

EXAMPLE 1.1

A three phase generator with rating 1000 kVA, 33 kV has its armature resistance and synchronous reactance as 20Ω/phase and 70Ω/phase. Calculate p.u. impedance of the generator.

SOLUTION

The generator ratings are chosen as base kV and base kVA.

$\therefore$ Base kilovolt, $kV_b = 33$ kV

Base kilovoltampere, $kVA_b = 1000$ kVA

Base impedance per phase: $Z_b = \frac{(kV_b)^2}{MVA_b} = \frac{(33)^2}{1000/1000} = 1089\ \Omega$

Actual impedance per phase: $Z = (20 + j70)\ \Omega$ / phase

$\therefore$ p.u. impedance, $Z_{pu} = \frac{\text{Actual impedance}}{\text{Base impedance}} = \frac{Z}{Z_b} = \frac{20 + j70}{1089} = 0.018 + j0.064$ p.u.

EXAMPLE 1.2

A three phase, Δ-Y transformer with rating 100 kVA, 11 kV/400 V has its primary and secondary leakage reactance as 12 Ω/phase and 0.05 Ω/phase respectively. Calculate the p.u. reactance of transformer.

SOLUTION

Case(i)

The high voltage winding (primary) ratings are chosen as base values.

$\therefore$ Base kilovolt, $kV_b = 11$ kV

Base kilovoltampere, $kVA_b = 100$ kVA

Base impedance per phase: $Z_b = \frac{(kV_b)^2}{MVA_b} = \frac{(11)^2}{100/1000} = 1210\ \Omega$

Transformer line voltage ratio, $K = \frac{400}{11{,}000} = 0.0364$

Total leakage reactance referred to primary: $X_{01} = X_1 + X_2' = X_1 + \frac{X_2}{K^2} = 12 + \frac{0.05}{(0.0364)^2} = 12 + 37.737$

$= 49.737\ \Omega$ / phase

p.u. reactance per phase, $X_{pu} = \frac{\text{Total leakage reactance}}{\text{Base impedance}}$

$$= \frac{X_{01}}{Z_b} = \frac{49.737}{1210} = 0.0411 \text{ p.u.}$$

Case(ii)

The low voltage winding (secondary) ratings are chosen as base values.

$\therefore$ Base kilovolt, $kV_b = 400/1000 = 0.4$ kV
Base kilovoltampere, $kVA_b = 100$ kVA

Base impedance per phase: $Z_b = \frac{(kV_b)^2}{MVA_b} = \frac{(0.4)^2}{100/1000} = 1.6\ \Omega$

Transformer line voltage ratio, $K = \frac{400}{11{,}000} = 0.0364$

Total leakage reactance referred to secondary: $X_{02} = X_1' + X_2 = K^2 X_1 + X_2$

$$= (0.0364)^2 \times 12 + 0.05 = 0.0159 + 0.05 = 0.0659\ \Omega/\text{phase}$$

p.u. reactance per phase, $X_{pu} = \frac{\text{Total leakage reactance}}{\text{Base impedance}} = \frac{X_{02}}{Z_b} = \frac{0.0659}{1.6} = 0.0411$ p.u.

Note:
1. It is observed that the p.u. reactance of a transformer referred to primary and secondary are same.
2. In three phase transformer if the voltage ratio K is obtained using line values then using this value of K the phase impedance per phase of star side can be directly transferred to delta side or vice-versa.

EXAMPLE 1.3

A three phase Y-Δ transformer is constructed using three identical single phase transformers of rating 200 kVA, 63.51 kV/11 kV transformer. The impedances of primary and secondary are 20 + j45Ω and 0.1 + j0.2Ω respectively. Calculate the p.u. impedance of the transformer.

SOLUTION

The three phase transformer is formed using three numbers of identical single phase transformers. Hence the kVA rating of three phase transformer is three times that of single phase transformer.

$\therefore$ kVA rating of three phase transformer $= 3 \times 200 = 600$ kVA

Line voltage rating of Y-Δ transformer $= 63.51 \times \sqrt{3}$ kV / 11 kV

$= 110$ kV / 11 kV

Case(i)

The high voltage winding (primary) ratings are chosen as base values.

$\therefore$ Base kilovolt, $kV_b = 110$ kV

Base kilovoltampere, $kVA_b = 600$ kVA

Base impedance per phase: $Z_b = \frac{(kV_b)^2}{MVA_b} = \frac{(110)^2}{600/1000} = 20166.7\ \Omega$

Transformer line voltage ratio, $K = \frac{11}{110} = 0.1$

Total impendace referred to primary: $Z_{01} = Z_1 + Z_2' = Z_1 + \frac{Z_2}{K^2} = 20 + j45 + \frac{0.1 + j0.2}{(0.1)^2} = 20 + j45 + 10 + j20$

$$= 30 + j65\ \Omega/\text{phase}$$

p.u. impedance per phase, $Z_{pu} = \frac{\text{Total impedance}}{\text{Base impedance}} = \frac{30 + j65}{20166.7} = 0.0015 + j0.0032$ p.u.

Case(ii)

The low voltage winding (secondary) ratings are chosen as base values.

$\therefore$ Base kilovolt, $kV_b = 11$ kV

Base kilovoltampere, $kVA_b = 600$ kVA

Base impedance per phase: $Z_b = \frac{(kV_b)^2}{MVA_b} = \frac{(11)^2}{600/1000} = 201.67\ \Omega$

Transformer line voltage ratio, $K = \frac{11}{110} = 0.1$

Total impedance referred to secondary: $Z_{02} = Z_1' + Z_2 = K^2 Z_1 + Z_2$

$$= (0.1)^2 \times (20 + j45) + 0.1 + j0.2 = 0.2 + j0.45 + 0.1 + j0.2$$

$$= 0.3 + j0.65\ \Omega/\text{phase}$$

p.u. reactance per phase, $Z_{pu} = \frac{\text{Total impedance}}{\text{Base impedance}} = \frac{0.3 + j0.65}{201.67} = 0.0015 + j0.0032$ p.u.

Note : 1. It is observed that the p.u. impedance of a transformer referred to primary and secondary are same.

2. In three phase transformer if the voltage ratio K is obtained using line values then using this value of K, the impedance per phase of star side can be directly transferred to delta side or vice-versa.

EXAMPLE 1.4

A 50 kW, three phase, Y connected load is fed by a 200 kVA transformer with voltage rating 11 kV/400 V through a feeder. The length of the feeder is 0.5 km and the impedance of the feeder is 0.1 + j0.2 Ω/km. If the load p.f. is 0.8, calculate the p.u. impedance of the load and feeder.

SOLUTION

Let us choose the secondary winding rating of transformer as base values.

$\therefore$ Base kilovolt, $kV_b = 400/1000 = 0.4$ kV

Base kilovoltampere, $kVA_b = 200$ kVA

Base impedance per phase: $Z_b = \frac{(kV_b)^2}{MVA_b} = \frac{(0.4)^2}{200/1000} = 0.8\ \Omega$

Actual impendace of feeder: $Z_{fed} = (0.1 + j0.2) \times 0.5 = 0.05 + j0.1\ \Omega$ / phase

p.u. impedance of feeder: $Z_{pu,\,fed} = \frac{\text{Actual impedance}}{\text{Base impedance}} = \frac{Z_{fed}}{Z_b}$

$= \frac{0.05 + j0.1}{0.8} = 0.0625 + j0.125$ p.u.

Given that P = 50 kW and pf = cosϕ = 0.8

$\therefore \sin\phi = \sin(\cos^{-1} 0.8) = 0.6$

Reactive power, $Q = \frac{P}{\cos\phi} \times \sin\phi = \frac{50}{0.8} \times 0.6 = 37.5$ kVAR

Load impedance per phase: $Z_L = \frac{|V_L|^2}{P - jQ} = \frac{400^2}{(50 - j37.5) \times 10^3} = \frac{400^2 \times 10^{-3}}{50 - j37.5}$

$= \frac{160}{62.5\angle -36.87^\circ} = 2.56\angle 36.87^\circ = 2.048 + j1.536$ W / phase

p.u value of load impedance: $Z_{L,pu} = \frac{\text{Load impedance}}{\text{Base impedance}} = \frac{Z_L}{Z_b} = \frac{2.048 + j1.536}{0.8} = 2.56 + j1.92$ p.u.

EXAMPLE 1.5

The three-phase ratings of a three winding transformer are

Primary : Y-connected, 110 kV, 20 MVA
Secondary : Y-connected, 13.2 kV, 15 MVA
Tertiary : Δ - connected, 2.1 kV, 0.5 MVA

Three short-circuit tests performed on this transformer yielded the following results (i) primary excited, secondary shorted : 2290 V, 52.5 A (ii) Primary excited, Tertiary shorted : 1785 V, 52.5 A (iii) Secondary excited, Tertiary shorted : 148 V, 328 A.

Find the p.u. impedances of the star-connected single-phase equivalent circuit for a base of 20 MVA, 110 kV in the primary circuit. Neglect resistances.

SOLUTION

Test (i) and (ii) are performed on primary winding, hence the p.u. impedances of Z_{ps} and Z_{pt} can be obtained directly using the primary winding ratings as base values.

The test (iii) is performed on secondary winding, hence the p.u. impedance Z_{st} is obtained using secondary winding rating as bases and then it can be converted to primary winding base.

To find p.u. value of Z_{ps} and Z_{pt}

Base kilovolt of primary circuit, $kV_{b,py} = 110$ kV

Base megavoltampere of primary circuit, $MVA_{b,py} = 20$ MVA

$\therefore$ Base impedance of primary circuit: $Z_{b,py} = \frac{(kV_{b,py})^2}{MVA_{b,py}} = \frac{(110)^2}{20} = 605\ \Omega$ / phase

The readings of test (i) can be used to calculate the value of Z_{ps} in Ω/phase.

Z_{ps} in Ω / phase $= \frac{2290/\sqrt{3}}{52.5} = 25.1835\ \Omega$ / phase

Note : The primary is star connected.

p.u. value of $Z_{ps} = \frac{Z_{ps} \text{ in } \Omega / \text{phase}}{Z_{b,py}} = \frac{25.1835}{605} = 0.0416$ p.u.

The readings of test (ii) can be used to calculate the value of Z_{pt} in Ω/phase.

Z_{pt} in Ω / phase $= \frac{1785/\sqrt{3}}{52.5} = 19.6299\ \Omega$ / phase

Note : The primary is star connected.

p.u. value of $Z_{pt} = \frac{Z_{pt} \text{ in } \Omega / \text{phase}}{Z_{b,py}} = \frac{19.6299}{605} = 0.0324$ p.u.

To find p.u. value of Z_{st}'

Base kilovolt of secondary circuit, $kV_{b,sec} = 13.2\ kV$

Base megavoltampere of secondary circuit, $MVA_{b,sec} = 15\ MVA$

$$\left.\begin{matrix}\therefore \text{Base impedance}\\ \text{of secondary circuit}\end{matrix}\right\} Z_{b,sec} = \frac{(kV_{b,sec})^2}{MVA_{b,sec}} = \frac{(13.2)^2}{15} = 11.616\ \Omega/\text{phase}$$

The readings of test (iii) can be used to calculate the value of Z_{st} in Ω/phase.

$$Z_{st} \text{ in } \Omega/\text{phase} = \frac{148/\sqrt{3}}{328} = 0.2605\ \Omega/\text{phase}$$

Note : The secondary is star connected.

$$\left.\begin{matrix}\text{p.u. value of } Z_{st} \text{ on the base}\\ \text{values of secondary circuit}\end{matrix}\right\} Z_{st} = \frac{Z_{st} \text{ in } \Omega/\text{phase}}{Z_{b,sec}} = \frac{0.2605}{11.616} = 0.0224 \text{ p.u.}$$

The p.u. value of Z_{st} on the secondary circuit base values can be converted to primary circuit base using the following formula.

$$Z_{pu,new} = Z_{pu,old} \times \left(\frac{kV_{b,old}}{kV_{b,new}}\right)^2 \times \left(\frac{MVA_{b,new}}{MVA_{b,old}}\right)$$

Here new refers to primary and old refers to secondary.

$$\left.\begin{matrix}\text{p.u. value of } Z_{st}' \text{ on the base}\\ \text{values of primary circuit}\end{matrix}\right\} Z_{st}' = 0.0224 \times \left(\frac{110}{110}\right)^2 \times \left(\frac{20}{15}\right) = 0.0299 \text{ p.u.}$$

Note : The $kV_{b,sec}$ when referred to primary side will be (110/13.2)×13.2 = 110 kV

To compute Z_p, Z_s' and Z_t'

$$Z_p = \frac{1}{2}\left[Z_{ps} + Z_{pt} - Z_{st}'\right] = \frac{1}{2}\left[0.0416 + 0.0324 - 0.0299\right] = 0.0221 \text{ p.u.}$$

$$Z_s' = \frac{1}{2}\left[Z_{ps} + Z_{st}' - Z_{pt}\right] = \frac{1}{2}\left[0.0416 + 0.0299 - 0.0324\right] = 0.0196 \text{ p.u.}$$

$$Z_t' = \frac{1}{2}\left[Z_{pt} + Z_{st}' - Z_{ps}\right] = \frac{1}{2}\left[0.0324 + 0.0299 - 0.0416\right] = 0.01035 \text{ p.u.}$$

The single phase star connected equivalent circuit of three winding transformer is shown in fig 1.5.1. Since the resistances are neglected the impedances are represented as pure reactances.

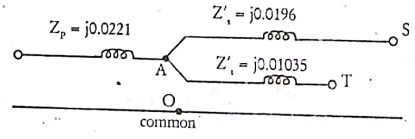


Fig 1.5.1 : Single phase equivalent circuit of three winding transformer

1.4 IMPEDANCE AND REACTANCE DIAGRAM

The Impedance or Reactance diagram of a power system is the equivalent circuit of the power system in which the various components of the power system are represented by their approximate or simplified equivalent circuits. This equivalent circuit of power system is used to analyse the performance of a system under load conditions (load flow studies) or to analyse the condition of the system under fault.

Impedance diagram

The impedance of a power system diagram is used for load flow studies. The impedance diagram can be obtained from the single line diagram by replacing all the components of the power system by their single phase equivalent circuit. The following approximations are made while forming impedance diagram.

1. *The current limiting impedances connected between the generator neutral and ground are neglected since under balanced conditions no current flows through neutral.*
2. *Since the magnetizing current of a transformer is very low when compared to load current the shunt branches in the equivalent circuit of the transformer can be neglected.*
3. *If the inductive reactance of a component is very high when compared to resistance then the resistance can be omitted, which introduces a little error in calculations.*

Reactance diagram

The reactance diagram is used for fault calculations. The following approximations are made in constructing reactance diagram, (when the system is balanced).

1. *The neutral to ground impedance of the generator is neglected for symmetrical faults.*
2. *Shunt branches in the equivalent circuits of transformer are neglected.*
3. *The resistances in the equivalent circuits of various components of the system are omitted.*
4. *All static loads are neglected.*
5. *Induction motors are neglected in computing fault current a few cycles after the fault occurs, because the current contributed by an induction motor dies out very quickly after the induction motor is short-circuited.*
6. *The capacitance of the transmission lines are neglected.*

The reactance diagram can be obtained from impedance diagram if we omit all static loads, all resistances, shunt branches of transformer and capacitance of transmission lines in the impedance diagram. The simplified representation of various components of power system in reactance diagram are shown in table 1.2.

The impedance and reactance diagram obtained from the above approximations are also called positive sequence impedance diagram and positive sequence reactance diagram.

Table-1.2 : Representation of components of power system in reactance diagram

Component	Equivalent circuit	Component	Equivalent circuit
3-phase generator	jX_g, I_g, E_g	Synchronous motor	jX_s, I_m, E_m
Transformer	% X	Transmission line	Inductive Reactance

A sample single-line diagram of a power system, its impedance and reactance diagrams are shown in fig 1.8.

The impedance or reactances of various components of power system in a single line diagram are expressed in percentage value or per unit calculated by taking their ratings as base values. When the impedance or reactance diagram is formed, all the impedances or reactances should be expressed in per unit calculated on a common base value. Hence it is necessary to convert all the p.u. reactances to a common base. The conversion of per unit reactances from one base to another can be performed using the following equation which is obtained from equation (1.16) after replacing Z by X.

$$X_{pu,new} = X_{pu,old} \times \left(\frac{KV_{b,old}}{KV_{b,new}}\right)^2 \times \left(\frac{MVA_{b,new}}{MVA_{b,old}}\right) \qquad \text{.....(1.47)}$$

Procedure to form reactance diagram from single line diagram

1. Select a base kilovoltampere or megavoltampere (kVA_b or MVA_b). The kVA_b or MVA_b will be same for all sections of the power system. In case of three phase power system, the kVA_b or MVA_b is three phase power rating.
2. Select a base kilovolt (kV_b) for one section of power system. In case of three phase power system, the kV_b is a line value. The various sections of power system works at different voltage levels and the voltage conversion is achieved by means of transformers. Hence the kV_b of one section of power system should be converted to a kV_b corresponding to another section using the transformer voltage ratio. In case of three phase transformer, line-to-line voltage ratio is used to transfer the kV_b on one section to another section.

$$kV_b \text{ on LT section} = kV_b \text{ on HT section} \times \frac{\text{LT voltage rating}}{\text{HT voltage rating}} \qquad \text{.....(1.48)}$$

$$kV_b \text{ on HT section} = kV_b \text{ on LT section} \times \frac{\text{HT voltage rating}}{\text{LT voltage rating}} \qquad \text{.....(1.49)}$$

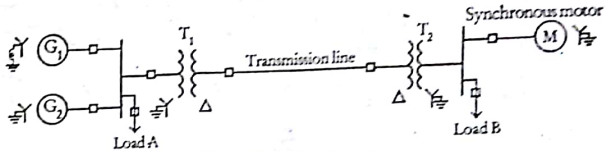


Fig a : Single line diagram

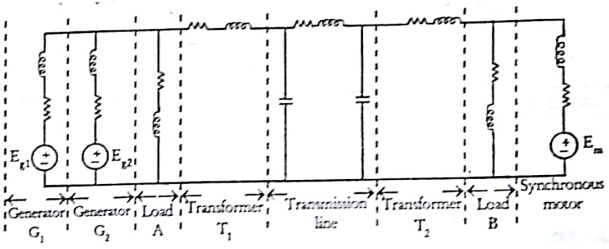


Fig b : Impedance diagram

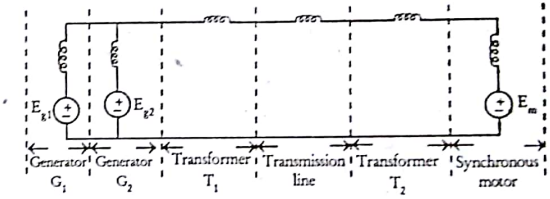


Fig c : Reactance diagram

Fig 1.8 : Single line diagram, impedance diagram and reactance diagram of a sample power system

3. *The impedance of the components of power system are expressed either in ohms (actual impedance) or in p.u. which is calculated using the component rating as base values. In reactance diagram, the resistances are neglected and the reactances of all the components are expressed on a common base. Hence starting from one end of power system the reactances of each component should be converted to p.u. reactances on the selected new base. When the specified reactance of the component is in ohms then*

$$\text{p.u. reactance} = \frac{\text{Actual reactance in ohms}}{\text{Base impedance}}$$

when the specified reactance of the component is in p.u. on the component rating as base values, then consider the component rating as old base values and selected base values as new bases. Now the p.u. reactance on new base can be calculated using the formula

$$X_{pu,new} = X_{pu,old} \times \left(\frac{kV_{b,old}}{kV_{b,new}}\right)^2 \times \left(\frac{MVA_{b,new}}{MVA_{b,old}}\right)$$

EXAMPLE 1.6

a) A generator is rated 500 MVA, 22 kV. Its Y-connected winding has a reactance of 1.1 p.u. Find the ohmic value of the reactance of winding.

b) If the generator is working in a circuit for which the bases are specified as 100 MVA, 20 kV. Then find the p.u. value of reactance of generator winding on the specified base.

SOLUTION

(a) The generator p.u. reactance will be specified by taking its rating as base values

$\therefore$ Base kilovolt, $kV_b = 22$ kV and Base megavoltampere, $MVA_b = 500$ MVA

$$\text{Base impedance, } Z_b = \frac{(kV_b)^2}{MVA_b} = \frac{22^2}{500} = 0.968\ \Omega$$

$$\text{Per unit reactance, } X_{pu} = \frac{\text{Actual reactance, } \Omega}{\text{Base impedance, } \Omega} = \frac{X}{Z_b}$$

$\therefore$ Actual reactance, $X = X_{pu} \times Z_b = 1.1 \times 0.968 = 1.0648\ \Omega$ / phase

(b) The formula used to convert the p.u. reactance specified on a base value to another base is given below

$$X_{pu,new} = X_{pu,old} \times \left(\frac{KV_{b,old}}{KV_{b,new}}\right)^2 \times \left(\frac{MVA_{b,new}}{MVA_{b,old}}\right)$$

The new base values are, $kV_{b,new} = 20$ kV and $MVA_{b,new} = 100$ MVA.
The old base values are $kV_{b,old} = 22$ kV and $MVA_{b,old} = 500$ MVA

$$\therefore X_{pu,new} = 1.1 \times \left(\frac{22}{20}\right)^2 \times \left(\frac{100}{500}\right) = 0.2662 \text{ p.u.}$$

EXAMPLE 1.7

A 300 MVA, 20 kV, 3ϕ generator has a subtransient reactance of 20%. The generator supplies 2 synchronous motors through a 64 km transmission line having transformers at both ends as shown in fig. 1.7.1. In this, T_1 is a 3ϕ transformer and T_2 is made of 3 single phase transformer of rating 100 MVA, 127/13.2 kV, 10% reactance. Series reactance of the transmission line is 0.5 Ω/km. Draw the reactance diagram with all the reactances marked in p.u. Select the generator rating as base values.

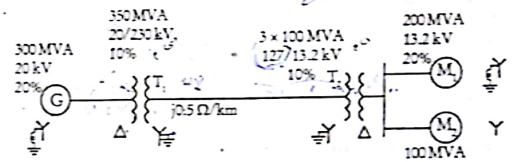


Fig 1.7.1

SOLUTION

Base megavoltampere, $MVA_{b,new} = 300$ MVA

Base kilovolt, $kV_{b,new} = 20$ kV

Reactance of Generator G

Since the generator rating and the base values are same, the generator p.u. reactance does not change.

$\therefore$ p.u. reactance of generator = 20% = 0.2 p.u.

Reactance of Transformer T_1

$$\left.\begin{matrix}\text{The new p.u. reactance}\\ \text{of Transformer } T_1\end{matrix}\right\} = X_{pu,old} \times \left(\frac{kV_{b,old}}{kV_{b,new}}\right)^2 \times \left(\frac{MVA_{b,new}}{MVA_{b,old}}\right)$$

$$= 0.1 \times \left(\frac{20}{20}\right)^2 \times \frac{300}{350} = 0.0857 \text{ p.u.}$$

Reactance of Transmission line

Reactance of transmission line = 0.5 Ω/km.

Total reactance of transmission line = $0.5 \times 64 = 32\ \Omega$.

$$\left.\begin{matrix}\text{Base kV on HT side}\\ \text{of transformer } T_1\end{matrix}\right\} = \text{Base kV on LT side} \times \frac{\text{HT voltage rating}}{\text{LT voltage rating}} = 20 \times \frac{230}{20} = 230 \text{ kV}$$

$$\text{Base impedance, } Z_b = \frac{(kV_b)^2}{MVA_b} = \frac{230^2}{300} = 176.33\ \Omega$$

$$\left.\begin{matrix}\text{Per unit reactance of}\\ \text{Transmission line}\end{matrix}\right\} = \frac{\text{Actual reactance}}{\text{Base impedance}} = \frac{32}{176.33} = 0.1815 \text{ p.u.}$$

Reactance of Transformer T_2

The transformer T_2 is a 3-phase transformer bank formed using three numbers of single phase transformers with voltage rating 127/13.2 kV. In this the high voltage side is star connected and low-voltage side is delta connected.

$$\therefore \text{Voltage ratio of line voltage of 3 - phase transformer bank} = \frac{\sqrt{3} \times 127}{13.2} = \frac{220}{13.2} \text{ kV}$$

$$\text{Base kV on LT side of transformer } T_2 = \text{Base kV on HT side} \times \frac{\text{LT voltage rating}}{\text{HT voltage rating}}$$

$$= 230 \times \frac{13.2}{220} = 13.8 \text{ kV}$$

$$\text{The new p.u. reactance of transformer } T_2 = X_{pu,\,old} \times \left(\frac{kV_{b,\,old}}{kV_{b,\,new}}\right)^2 \times \left(\frac{MVA_{b,\,new}}{MVA_{b,\,old}}\right)$$

$$= 0.1 \times \left(\frac{13.2}{13.8}\right)^2 \times \left(\frac{300}{3 \times 100}\right) = 0.0915 \text{ p.u.}$$

Reactance of M_1

$$\text{p.u. reactance of } M_1 \text{ on new base} = X_{pu,\,old} \times \left(\frac{kV_{b,\,old}}{kV_{b,\,new}}\right)^2 \left(\frac{MVA_{b,\,new}}{MVA_{b,\,old}}\right)$$

$$= 0.2 \times \left(\frac{13.2}{13.8}\right)^2 \times \frac{300}{200} = 0.2745 \text{ p.u.}$$

Reactance of M_2

$$\text{p.u. reactance of } M_2 \text{ on new base} = X_{pu,\,old} \times \left(\frac{kV_{b,\,old}}{kV_{b,\,new}}\right)^2 \left(\frac{MVA_{b,\,new}}{MVA_{b,\,old}}\right)$$

$$= 0.2 \times \left(\frac{13.2}{13.8}\right)^2 \times \frac{300}{100} = 0.549 \text{ p.u.}$$

Reactance diagram

The reactance diagram is shown in fig 1.7.2.

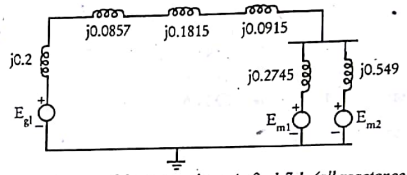


Fig 1.7.2 : Reactance diagram of the system shown in fig 1.7.1. (all reactance values are in p.u.)

EXAMPLE 1.8

A 120 MVA, 19.5 kV generator has a synchronous reactance of 0.15 p.u. and it is connected to a transmission line through a transformer rated 150 MVA, 230/18 kV (Y/Δ) with X = 0.1 pu.

(a) Calculate the p.u. reactances by taking generator rating as base values.

(b) Calculate the p.u. reactance by taking transformer rating as base values.

(c) Calculate the p.u. reactances for a base value of 100 MVA and 220 kV on HT side of transformer.

SOLUTION

(a) Base megavoltampere, $MVA_{b,\,new} = 120$ MVA

Base kilovolt, $kV_{b,\,new} = 19.5$ kV

Since the generator ratings are chosen as base values, its p.u. reactance will not change.

$\therefore$ Reactance of generator = 0.15 p.u.

$$\text{New p.u. reactance of transformer} = X_{pu,\,old} \times \left(\frac{kV_{b,\,old}}{kV_{b,\,new}}\right)^2 \times \left(\frac{MVA_{b,\,new}}{MVA_{b,\,old}}\right)$$

$$= 0.1 \times \left(\frac{18}{19.5}\right)^2 \times \frac{120}{150} = 0.0682 \text{ p.u.}$$

(b) Base megavoltampere, $MVA_{b,\,new} = 150$ MVA

Base kilovolt, $kV_{b,\,new} = 18$ kV

Since the transformer ratings are chosen as base values, its pu reactance will not change.

$\therefore$ Reactance of transformer = 0.1 p.u.

$$\text{New p.u. reactance of generator} = X_{pu,\,old} \times \left(\frac{kV_{b,\,old}}{kV_{b,\,new}}\right)^2 \times \left(\frac{MVA_{b,\,new}}{MVA_{b,\,old}}\right)$$

$$= 0.15 \times \left(\frac{19.5}{18}\right)^2 \times \frac{150}{120} = 0.22 \text{ p.u.}$$

(c) Base megavoltampere, $MVA_{b,new} = 100\,MVA$

Base kilovolt, $kV_{b,new} = 220\,kV$

In this case the base values are neither generator ratings nor transformer ratings, Hence both the p.u. reactances should be converted to new base.

$$\text{New p.u. reactance of transformer} = X_{pu,\,old} \times \left(\frac{kV_{b,\,old}}{kV_{b,new}}\right)^2 \times \left(\frac{MVA_{b,\,new}}{MVA_{b,\,old}}\right)$$

$$= 0.1 \times \left(\frac{230}{220}\right)^2 \times \frac{100}{150} = 0.0729 \text{ p.u.}$$

The generator is connected to LT side of transformer.

$$\left.\begin{array}{l}\therefore \text{Base kV referred of} \\ \text{LT side of transformer}\end{array}\right\} = \text{Base kV on HT side} \times \frac{\text{LT voltage rating}}{\text{HT voltage rating}}$$

$$= 220 \times \frac{18}{230} = 17.22 \text{ KV}$$

Now, $kV_{b,new} = 17.22$ kV

$$\left.\begin{array}{l}\text{New p.u. reactance} \\ \text{of generator}\end{array}\right\} = X_{pu,\,old} \times \left(\frac{kV_{b,\,old}}{kV_{b,\,new}}\right)^2 \times \left(\frac{MVA_{b,\,new}}{MVA_{b,\,old}}\right)$$

$$= 0.15 \times \left(\frac{19.5}{17.22}\right)^2 \times \left(\frac{100}{120}\right) = 0.1603 \text{ p.u.}$$

RESULT

a $kV_b = 19.5\,kV$; $MVA_b = 120\,MVA$, $X_{pu,gen} = 0.15$ p.u. ; $X_{pu,tr} = 0.0682$ p.u.

b $kV_b = 18\,kV$; $MVA_b = 150\,MVA$, $X_{pu,gen} = 0.22$ p.u. ; $X_{pu,tr} = 0.1$ p.u.

c $kV_b = 220\,kV$; $MVA_b = 100\,MVA$, $X_{pu,gen} = 0.1603$ p.u. ; $X_{pu,tr} = 0.0729$ p.u.

EXAMPLE 1.9

The single line diagram of an unloaded power system is shown in fig 1.9.1. The generator and transformers are rated as follows.

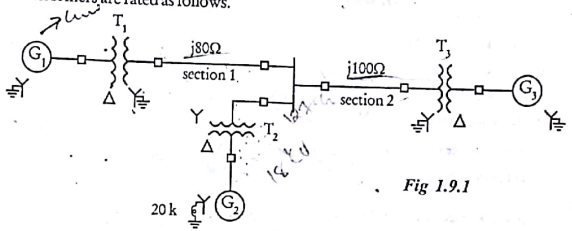


Fig 1.9.1

Generator, $G_1 = 20\,MVA$, 13.8 kV, X" = 20%

Generator, $G_2 = 30\,MVA$, 18 kV, X" = 20%

Generator, $G_3 = 30\,MVA$, 20 kV, X" = 20%

Transformer, $T_1 = 25\,MVA$, 220/13.8 kV, X = 10%

Transformer, $T_2 = 3$ single phase units each rated at 10 MVA, 127/18 kV, X = 10%.

Transformer, $T_3 = 35\,MVA$, 220/22 kV, X = 10%.

Draw the reactance diagram using a base of 50 MVA and 13.8 kV on the generator G_1.

SOLUTION

Base megavoltampere, $MVA_{b,new} = 50\,MVA$

Base kilovolt, $kV_{b,new} = 13.8\,kV$

Reactance of Generator G_1

$$\left.\begin{array}{l}\text{New p.u. reactance} \\ \text{of generator } G_1\end{array}\right\} = X_{pu,\,old} \times \left(\frac{kV_{b,\,old}}{kV_{b,\,new}}\right)^2 \times \left(\frac{MVA_{b,\,new}}{MVA_{b,\,old}}\right)$$

$$= 0.2 \times \left(\frac{13.8}{13.8}\right)^2 \times \left(\frac{50}{20}\right) = 0.5 \text{ p.u.}$$

Reactance of Transformer T_1

$$\left.\begin{array}{l}\text{New p.u. reactance} \\ \text{of transformer } T_1\end{array}\right\} = X_{pu,\,old} \times \left(\frac{kV_{b,\,old}}{kV_{b,\,new}}\right)^2 \times \left(\frac{MVA_{b,\,new}}{MVA_{b,\,old}}\right) = 0.1 \times \left(\frac{13.8}{13.8}\right)^2 \times \left(\frac{50}{25}\right) = 0.2 \text{ p.u.}$$

Reactance of Transmission lines

$$\left.\begin{matrix}\text{Base kV on HT side}\\ \text{of transformer } T_1\end{matrix}\right\} = \text{Base kV on LT side} \times \frac{\text{HT voltage rating}}{\text{LT voltage rating}} = 13.8 \times \frac{220}{13.8} = 220 \text{ kV}$$

Now, $kV_{b,\,new} = 220$ kV

$$\left.\begin{matrix}\text{Base impedance on HT}\\ \text{side of transformer}\end{matrix}\right\} = \frac{(kV_{b,\,new})^2}{MVA_{b,\,new}} = \frac{220^2}{50} = 968\ \Omega$$

$$\left.\begin{matrix}\text{p.u. reactance of section 1}\\ \text{of transmission line}\end{matrix}\right\} = \frac{\text{Actual reactance, } \Omega}{\text{Base impedance, } \Omega} = \frac{80}{968} = 0.0826 \text{ p.u.}$$

$$\left.\begin{matrix}\text{p.u. reactance of section 2}\\ \text{of transmission line}\end{matrix}\right\} = \frac{\text{Actual reactance, } \Omega}{\text{Base impedance, } \Omega} = \frac{100}{968} = 0.1033 \text{ p.u.}$$

Reactance of Transformer T_2

The transformer T_2 is a 3-phase transformer bank formed using three numbers of single phase transformer with voltage rating 127/18 kV. In this the HT side is star connected and LT side is delta connected.

$$\left.\begin{matrix}\therefore \text{ Voltage ratio of line voltage}\\ \text{of 3-phase transformer bank}\end{matrix}\right\} = \frac{\sqrt{3} \times 127}{18} = \frac{220}{18} \text{ kV}$$

$$\left.\begin{matrix}\text{Base kV on LT side}\\ \text{of transformer } T_2\end{matrix}\right\} = \text{Base kV on HT side} \times \frac{\text{LT voltage rating}}{\text{HT voltage rating}} = 220 \times \frac{18}{220} = 18 \text{ kV}$$

Now, $kV_{b,\,new} = 18$ kV

$$\left.\begin{matrix}\text{New p.u. reactance}\\ \text{of transformer } T_2\end{matrix}\right\} = X_{pu,\,old} \times \left(\frac{kV_{b,\,old}}{kV_{b,\,new}}\right)^2 \times \left(\frac{MVA_{b,\,new}}{MVA_{b,\,old}}\right)$$

$$= 0.1 \times \left(\frac{18}{18}\right)^2 \times \left(\frac{50}{3 \times 10}\right) = 0.1667 \text{ p.u.}$$

Reactance of Generator G_2

$$\left.\begin{matrix}\text{New p.u. reactance}\\ \text{of generator } G_2\end{matrix}\right\} = X_{pu,\,old} \times \left(\frac{kV_{b,\,old}}{kV_{b,\,new}}\right)^2 \times \left(\frac{MVA_{b,\,new}}{MVA_{b,\,old}}\right)$$

$$= 0.2 \times \left(\frac{18}{18}\right)^2 \times \left(\frac{50}{30}\right) = 0.3333 \text{ p.u.}$$

Reactance of Transformer T_3

$$\left.\begin{matrix}\text{Base kV on LT side}\\ \text{of transformer } T_3\end{matrix}\right\} = \text{Base kV on HT side} \times \frac{\text{LT voltage rating}}{\text{HT voltage rating}} = 220 \times \frac{22}{220} = 22 \text{ kV}$$

Now, $kV_{b,\,new} = 22$ kV

$$\left.\begin{matrix}\text{New p.u. reactance}\\ \text{of transformer } T_3\end{matrix}\right\} = X_{pu,\,old} \times \left(\frac{kV_{b,\,old}}{kV_{b,\,new}}\right)^2 \times \left(\frac{MVA_{b,\,new}}{MVA_{b,\,old}}\right) = 0.1 \times \left(\frac{22}{22}\right)^2 \times \left(\frac{50}{35}\right) = 0.1429 \text{ p.u.}$$

Reactance of Generator G_3

$$\left.\begin{matrix}\text{New p.u. reactance}\\ \text{of generator } G_3\end{matrix}\right\} = X_{pu,\,old} \times \left(\frac{kV_{b,\,old}}{kV_{b,\,new}}\right)^2 \times \left(\frac{MVA_{b,\,new}}{MVA_{b,\,old}}\right)$$

$$= 0.2 \times \left(\frac{20}{22}\right)^2 \times \left(\frac{50}{30}\right) = 0.2755 \text{ p.u.}$$

Reactance diagram

The reactance diagram is shown in fig 1.9.2.

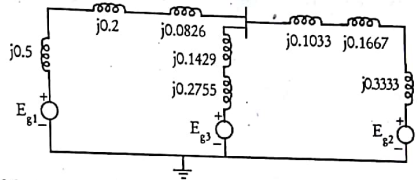


Fig 1.9.2 : Reactance diagram of the system shown in fig 1.9.1 (all reactance values are in p.u.)

EXAMPLE 1.10

Draw the reactance diagram for the power system shown in fig 1.10.1. Neglect resistance and use a base of 100 MVA, 220 kV in 50 Ω line. The ratings of the generator, motor and transformer are given below.

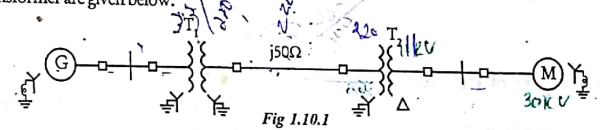


Fig 1.10.1

Generator : 40 MVA, 25 kV, X″ = 20%

Synchronous motor : 50 MVA, 11 kV, X″ = 30%

Y-Y Transformer : 40 MVA, 33/220 kV, X = 15%

Y-Δ Transformer : 30 MVA, 11/220 kV (Δ/Y), X = 15%

SOLUTION

Base megavoltampere, $MVA_{b,new} = 100\,MVA$

Base kilovolt, $kV_{b,new} = 220\,kV$

Reactance of Transmission line

$$\text{Base impedance} = \frac{(kV_{b,new})^2}{MVA_{b,new}} = \frac{220^2}{100} = 484\ \Omega$$

$$\left.\begin{matrix}\text{p.u. reactance of}\\ \text{transmission line}\end{matrix}\right\} = \frac{\text{Actual reactance, }\Omega}{\text{Base impedance, }\Omega} = \frac{50}{484} = 0.1033\ \text{p.u.}$$

Reactance of Transformer T_1

$$\left.\begin{matrix}\text{Base kV on LT side}\\ \text{of Transformer } T_1\end{matrix}\right\} = \text{Base kV on HT side} \times \frac{\text{LT voltage rating}}{\text{HT voltage rating}} = 220 \times \frac{33}{220} = 33\ \text{kV}$$

Now, $kV_{b,new} = 33\ kV$

$$\left.\begin{matrix}\text{New p.u. reactance of}\\ \text{Transformer } T_1\end{matrix}\right\} = X_{pu,old} \times \left(\frac{kV_{b,old}}{kV_{b,new}}\right)^2 \times \left(\frac{MVA_{b,new}}{MVA_{b,old}}\right)$$

$$= 0.15 \times \left(\frac{33}{33}\right)^2 \times \left(\frac{100}{40}\right) = 0.375\ \text{p.u.}$$

Reactance of Generator G

$$\left.\begin{matrix}\text{New p.u. reactance}\\ \text{of generator G}\end{matrix}\right\} = X_{pu,old} \times \left(\frac{kV_{b,old}}{kV_{b,new}}\right)^2 \times \left(\frac{MVA_{b,new}}{MVA_{b,old}}\right)$$

$$= 0.2 \times \left(\frac{25}{33}\right)^2 \times \left(\frac{100}{40}\right) = 0.287\ \text{p.u.}$$

Reactance of Transformer T_2

$$\left.\begin{matrix}\text{Base kV on LT side}\\ \text{of Transformer } T_2\end{matrix}\right\} = \text{Base kV on HT side} \times \frac{\text{LT voltage rating}}{\text{HT voltage rating}} = 220 \times \frac{11}{220} = 11\ \text{kV}$$

Now, $kV_{b,new} = 11\ kV$

$$\left.\begin{matrix}\text{New p.u. reactance of}\\ \text{Transformer } T_2\end{matrix}\right\} = X_{pu,old} \times \left(\frac{kV_{b,old}}{kV_{b,new}}\right)^2 \times \left(\frac{MVA_{b,new}}{MVA_{b,old}}\right)$$

$$= 0.15 \times \left(\frac{11}{11}\right)^2 \times \left(\frac{100}{30}\right) = 0.5\ \text{p.u.}$$

Reactance of Synchronous motor

$$\left.\begin{matrix}\text{New p.u. reactance of}\\ \text{synchronous motor}\end{matrix}\right\} = X_{pu,old} \times \left(\frac{kV_{b,old}}{kV_{b,new}}\right)^2 \times \left(\frac{MVA_{b,new}}{MVA_{b,old}}\right)$$

$$= 0.3 \times \left(\frac{11}{11}\right)^2 \times \left(\frac{100}{50}\right) = 0.6\ \text{p.u.}$$

Reactance diagram

The reactance diagram is shown in fig 1.10.2.

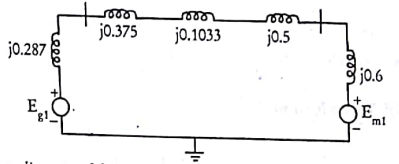


Fig 1.10.2 : Reactance diagram of the system shown in fig 1.10.1 (all reactance values are in p.u.)

EXAMPLE 1.11

Draw the reactance diagram for the power system shown in fig 1.11.1. The ratings of generator, motor and transformers are given below. Neglect resistance and use a base of 50 MVA, 138 kV in the 40 Ω line.

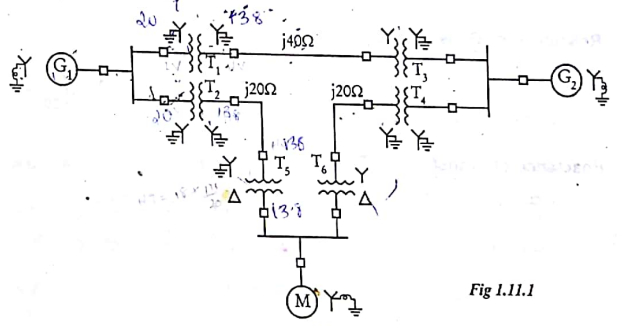


Fig 1.11.1

Generator G_1 : 20 MVA, 18 kV, $X'' = 20\%$
Generator G_2 : 20 MVA, 18 kV, $X'' = 20\%$
Synchronous motor : 30 MVA, 13.8 kV, $X'' = 20\%$
3-phase, Y-Y Transformer : 20 MVA, 138/20 kV, $X = 10\%$
3-phase, Y-Δ Transformer : 15 MVA, 138/13.8 kV, $X = 10\%$

SOLUTION

Base megavoltampere, $MVA_{b,new} = 50$ MVA
Base kilovolt, $kV_{b,new} = 138$ kV

Reactance of j40Ω Transmission line

$$\text{Base impedance} = \frac{(kV_{b,new})^2}{MVA_{b,new}} = \frac{138^2}{50} = 380.88\ \Omega$$

$$\left.\begin{matrix}\text{p.u. reactance of}\\ 40\Omega \text{ transmission line}\end{matrix}\right\} = \frac{\text{Actual reactance, } \Omega}{\text{Base impedance, } \Omega} = \frac{40}{380.88} = 0.105 \text{ p.u.}$$

Reactance of Transformer T_1

$$\left.\begin{matrix}\text{Base kV on LT side}\\ \text{of Transformer } T_1\end{matrix}\right\} = \text{Base kV on HT side} \times \frac{\text{LT voltage rating}}{\text{HT voltage rating}} = 138 \times \frac{20}{138} = 20 \text{ kV}$$

Now, $kV_{b,new} = 20$ kV

$$\left.\begin{matrix}\text{New p.u. reactance of}\\ \text{Transformer } T_1\end{matrix}\right\} = X_{pu,old} \times \left(\frac{kV_{b,old}}{kV_{b,new}}\right)^2 \times \left(\frac{MVA_{b,new}}{MVA_{b,old}}\right) = 0.1 \times \left(\frac{20}{20}\right)^2 \times \left(\frac{50}{20}\right) = 0.25 \text{ p.u.}$$

Reactance of Generator G_1

$$\left.\begin{matrix}\text{New p.u. reactance}\\ \text{of generator } G_1\end{matrix}\right\} = X_{pu,old} \times \left(\frac{kV_{b,old}}{kV_{b,new}}\right)^2 \times \left(\frac{MVA_{b,new}}{MVA_{b,old}}\right) = 0.2 \times \left(\frac{18}{20}\right)^2 \times \left(\frac{50}{20}\right) = 0.405 \text{ p.u.}$$

Reactance of Transformer T_2

$$\left.\begin{matrix}\text{New p.u. reactance of}\\ \text{Transformer } T_2\end{matrix}\right\} = X_{pu,old} \times \left(\frac{kV_{b,old}}{kV_{b,new}}\right)^2 \times \left(\frac{MVA_{b,new}}{MVA_{b,old}}\right)$$

$$= 0.1 \times \left(\frac{20}{20}\right)^2 \times \left(\frac{50}{20}\right) = 0.25 \text{ p.u.}$$

Reactance of j20Ω Transmission line

$$\left.\begin{matrix}\text{Base kV on HT side}\\ \text{of Transformer } T_2\end{matrix}\right\} = \text{Base kV on LT side} \times \frac{\text{HT voltage rating}}{\text{LT voltage rating}}$$

$$= 20 \times \frac{138}{20} = 138 \text{ kV}$$

Now, $kV_{b,new} = 138$ kV

$$\text{Base impedance} = \frac{kV_b^2}{MVA_b} = \frac{138^2}{50} = 380.88\Omega$$

$$\left.\begin{matrix}\text{p.u. reactance of } 20\Omega\\ \text{Transmission line}\end{matrix}\right\} = \frac{\text{Actual reactance, } \Omega}{\text{Base impedance, } \Omega} = \frac{20}{380.88} = 0.0525 \text{ p.u.}$$

Here it is observed that both the sections of j20Ω transmission lines have same values of reactances and base kV's. Hence their p.u. reactances will be same.

Reactance of Transformer T_5

$$\left.\begin{matrix}\text{New p.u. reactance of}\\ \text{Transformer } T_5\end{matrix}\right\} = X_{pu,old} \times \left(\frac{kV_{b,old}}{kV_{b,new}}\right)^2 \times \left(\frac{MVA_{b,new}}{MVA_{b,old}}\right)$$

$$= 0.1 \times \left(\frac{138}{138}\right)^2 \times \left(\frac{50}{15}\right) = 0.333 \text{ p.u.}$$

Reactance of Synchronous motor

$$\left.\begin{matrix}\text{Base kV on LT side}\\ \text{of Transformer } T_2\end{matrix}\right\} = \text{Base kV on HT side} \times \frac{\text{LT voltage rating}}{\text{HT voltage rating}} = 138 \times \frac{13.8}{138} = 13.8 \text{ kV}$$

Now, $kV_{b,new} = 13.8$ kV

$$\left.\begin{matrix}\text{New p.u. reactance}\\ \text{of synchronous motor}\end{matrix}\right\} = X_{pu,old} \times \left(\frac{kV_{b,old}}{kV_{b,new}}\right)^2 \times \left(\frac{MVA_{b,new}}{MVA_{b,old}}\right)$$

$$= 0.2 \times \left(\frac{13.8}{13.8}\right)^2 \times \left(\frac{50}{30}\right) = 0.333 \text{ p.u.}$$

Reactances of T_6, T_4, T_3 and G_2

1. *The transformer T_6 is identical to that of T_5. Hence p.u. reactance of T_5 and T_6 are same.*
2. *The transformers T_1, T_2, T_3 and T_4 are identical. Hence their p.u. reactances are same.*
3. *The generator G_2 is identical to that of G_1. Hence their p.u. reactances are same.*

Reactance diagram

The reactance diagram of the power system is shown in fig 1.11.2.

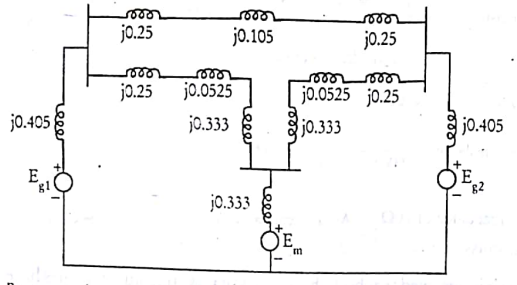


Fig 1.11.2 : Reactance diagram of the system shown in fig 1.11.1 (all reactance values are in p.u.)

EXAMPLE 1.12

A 15 MVA, 8.5 kV, 3-phase generator has a subtransient reactance of 20%. It is connected through a Δ-Y transformer to a high voltage transmission line having a total series reactance of 70 ohms. The load end of the line has Y-Y step down transformer. Both transformer banks are composed of single phase transformers connected for 3-phase operation. Each of three transformers composing three phase bank is rated 6667 kVA, 10/100 kV, with a reactance of 10%. The load represented as impedance, is drawing 10 MVA at 12.5 kV and 0.8 pf lagging. Draw the single line diagram of the power network. Choose a base of 10 MVA, 12.5 kV in the load circuit and determine the reactance diagram. Determine also the voltage at the terminals of the generator.

SOLUTION

The single line diagram of the power system is shown in fig 1.12.1.

Base values

Base megavoltampere, $MVA_{b,new} = 10\,MVA$

Base kilovolt, $kV_{b,new} = 12.5\,kV$

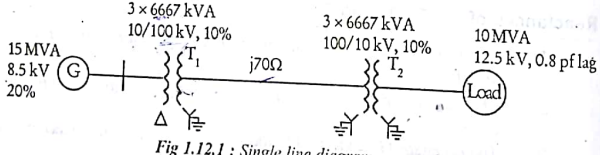


Fig 1.12.1 : Single line diagram

Reactance of Transformer T_2

Voltage ratio of line voltage of transformer $T_2 = \dfrac{100 \times \sqrt{3}\,kV}{10 \times \sqrt{3}\,kV} = \dfrac{173.2\,kV}{17.32\,kV}$

3-phase kVA rating of transformer $T_2 = 3 \times 6667 = 20{,}000\,kVA = 20\,MVA$

$\therefore kV_{b,old} = 17.32\,kV$ (on LT side)

$MVA_{b,old} = 20\,MVA$

$$\left.\begin{matrix}\text{New p.u. reactance}\\ \text{of transformer } T_2\end{matrix}\right\} = X_{pu,old} \times \left(\frac{kV_{b,old}}{kV_{b,new}}\right)^2 \times \left(\frac{MVA_{b,new}}{MVA_{b,old}}\right)$$

$$= 0.1 \times \left(\frac{17.32}{12.5}\right)^2 \times \left(\frac{10}{20}\right) = 0.096\ \text{p.u.}$$

Reactance of Transmission line

$$\left.\begin{matrix}\text{Base kV on HT side}\\ \text{of Transformer } T_2\end{matrix}\right\} = \text{Base kV on LT side} \times \frac{\text{HT voltage rating}}{\text{LT voltage rating}} = 12.5 \times \frac{173.2}{17.32} = 125\ kV$$

Now, $kV_{b,new} = 125\,kV$

$$\text{Base impedance } Z_b = \frac{(kV_{b,new})^2}{MVA_{b,new}} = \frac{(125)^2}{10} = 1562.5\,\Omega$$

$$\left.\begin{matrix}\text{p.u. reactance of}\\ \text{Transmission line}\end{matrix}\right\} = \frac{\text{Actual impedance, }\Omega}{\text{Base impedance, }\Omega} = \frac{70}{1562.5} = 0.0448\ \text{p.u.}$$

Reactance of Transformer T_1

Voltage ratio of line voltage of transformer $T_1 = \dfrac{10\,kV}{100 \times \sqrt{3}\,kV} = \dfrac{10\,kV}{173.2\,kV}$

3-phase kVA rating of transformer $T_1 = 3 \times 6667 = 20{,}000\,kVA = 20\,MVA$

$\therefore kV_{b,old} = 173.2\,kV$ (on HT side) $kV_{b\,new} = 125\,kV$

$MVA_{b,old} = 20\,MVA$

$$\left.\begin{matrix}\text{New p.u. reactance}\\ \text{of transformer } T_1\end{matrix}\right\} = X_{pu,old} \times \left(\frac{kV_{b,old}}{kV_{b,new}}\right)^2 \times \left(\frac{MVA_{b,new}}{MVA_{b,old}}\right)$$

$$= 0.1 \times \left(\frac{173.2}{125}\right)^2 \times \left(\frac{10}{20}\right) = 0.096\ \text{p.u.}$$

Reactance of Generator

$$\left.\begin{matrix}\text{Base kV on LT side}\\ \text{of Transformer } T_1\end{matrix}\right\} = \text{Base kV on HT side} \times \frac{\text{LT voltage rating}}{\text{HT voltage rating}} = 125 \times \frac{10}{173.2} = 7.217 \text{ kV}$$

Now, $kV_{b,new} = 7.217$ kV

$$\left.\begin{matrix}\text{New p.u. reactance}\\ \text{of generator}\end{matrix}\right\} = X_{pu,old} \times \left(\frac{kV_{b,old}}{kV_{b,new}}\right)^2 \times \left(\frac{MVA_{b,new}}{MVA_{b,old}}\right)$$

$$= 0.2 \times \left(\frac{8.5}{7.217}\right)^2 \times \left(\frac{10}{15}\right) = 0.185 \text{ p.u.}$$

Load

This can be represented as constant current load.

p.f of load = 0.8 lag

$\therefore$ p.f. angle = $-\cos^{-1} 0.8 = -36.87°$

Complex load power = $10 \angle -36.87°$ MVA

$$\text{p.u. value of load (power)} = \frac{\text{Actual load MVA}}{\text{Base value of MVA}} = \frac{10 \angle -36.87°}{10} = 1 \angle -36.87° \text{ p.u.}$$

$$\text{p.u. value of load voltage} = \frac{\text{Actual load voltage}}{\text{Base voltage}} = \frac{12.5 \text{ kV}}{12.5 \text{ kV}} = 1.0 \text{ p.u.}$$

Let, I = Load current in p.u.

V = Load voltage in p.u.

We know that, V × I = p.u. value of load (power)

$$\therefore I = \frac{\text{p.u. value of load}}{V} = \frac{1 \angle -36.87°}{1.0} = 1 \angle -36.87°$$

Reactance diagram

The reactance diagram of the system is shown in fig 1.12.2.

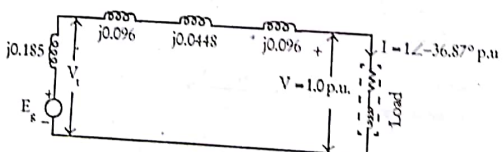


Fig 1.12.2 : Reactance diagram of the system shown in fig 1.12.1 (all reactance values are in p.u.)

To find the terminal voltage of the generator

With reference to fig 1.12.2,

The terminal voltage of the generator, $V_1 = V + I(j0.096 + j0.0448 + j0.096)$

$= 1.0 + 1\angle -36.87° (j0.2368)$

$= 1.0 + 1\angle -36.87° \times 0.2368\angle 90°$

$= 1.0 + 0.2368\angle 53.13°$

$= 1.0 + 0.1421 + j0.1894$

$= 1.1421 + j0.1894$

$= 1.1577\angle 9.4°$ p.u.

$$\left.\begin{matrix}\text{Actual value of generator}\\ \text{terminal voltage}\end{matrix}\right\} = \text{p.u. value of voltage} \times \left(\begin{matrix}\text{Base kV on}\\ \text{LT side of}\\ \text{transformer } T_1\end{matrix}\right)$$

$= 1.1577\angle 9.4° \times 7.217 = 8.355\angle 9.4°$ kV

1.5 NODE EQUATIONS AND BUS ADMITTANCE MATRIX

The meeting point of various components in a power system is called bus. (practically, the element used to connect one component to another is called bus). The bus or bus bar is a conductor made of copper or aluminium having negligible resistance. Hence the bus bar will have zero voltage drop when it conducts the rated current) Therefore the buses are considered as points of constant voltage in a power system.

When the power system is represented by impedance / reactance diagram, it can be considered as a circuit or network. The buses can be treated as nodes and the voltages of all buses (nodes) can be solved by conventional node analysis technique.

Let N be the number of major or principal nodes in a circuit or network. (The principal nodes are meeting points of more than two elements). Since the voltage of a node can be measured only with respect to a reference point, one of the node is considered as reference node. Now the network will have (N-1) independent voltages. In nodal analysis, the independent voltages are solved by writing Kirchoff's Current Law (KCL) equation for (N-1) nodes in the circuit. For writing KCL equations, the voltage sources in the circuit should be converted to equivalent current sources.

Let, $V_1, V_2, V_3, \ldots, V_n$ = Node voltages of nodes 1, 2, 3 ....., n respectively.

$I_{11}, I_{22}, I_{33}, \ldots, I_{nn}$ = Sum of current sources connected (or injecting current) to nodes 1, 2, 3 ....., n respectively.

Y_j = Sum of admittances connected to node-j.

Y_{jk} = Negative of sum of admittances connected between node-j and node-k.

Note : If the direction of current in current source is towards the node then the source is considered positive and if it is away from the node then the source is negative.

Now the n-number of nodal equations for N-bus system will be in the form shown below (Here n = N-1)

$$Y_{11}V_1 - Y_{12}V_2 - Y_{13}V_3 - \text{.....} - Y_{1n}V_n = I_{11}$$
$$Y_{21}V_1 - Y_{22}V_2 - Y_{23}V_3 - \text{.....} - Y_{2n}V_n = I_{22}$$
$$Y_{31}V_1 - Y_{32}V_2 - Y_{33}V_3 - \text{.....} - Y_{3n}V_n = I_{33}$$
$$\vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots$$
$$Y_{n1}V_1 - Y_{n2}V_2 - Y_{n3}V_3 - \text{.....} - Y_{nn}V_n = I_{nn}$$

The above n-number of equations can be arranged in the matrix form as shown in equ(1.50).

$$\begin{bmatrix} Y_{11} & Y_{12} & Y_{13} & \text{.....} & Y_{1n} \\ Y_{21} & Y_{22} & Y_{23} & \text{.....} & Y_{2n} \\ Y_{31} & Y_{32} & Y_{33} & \text{.....} & Y_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ Y_{n1} & Y_{n2} & Y_{n3} & \text{.....} & Y_{nn} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ \vdots \\ V_n \end{bmatrix} = \begin{bmatrix} I_{11} \\ I_{22} \\ I_{33} \\ \vdots \\ I_{nn} \end{bmatrix} \quad \text{.....(1.50)}$$

In matrix notation the equ(1.50) can be written as,

$$YV = I \quad \text{.....(1.51)}$$

In power system the Y-matrix is designated as Y_{bus} and called bus admittance matrix. The node voltages are called bus voltages.

Therefore the equ(1.51) can be written as shown in equ(1.52)

$$Y_{bus}V = I \quad \text{.....(1.52)}$$

where, Y_{bus} = Bus admittances matrix of order $(n \times n)$
V = Bus voltage matrix of order $(n \times 1)$
I = Currents sources matrix of order $(n \times 1)$
n = Number of independent buses in the system.

$$\text{Bus admittance matrix, } Y_{bus} = \begin{bmatrix} Y_{11} & Y_{12} & Y_{13} & \text{.....} & Y_{1n} \\ Y_{21} & Y_{22} & Y_{23} & \text{.....} & Y_{2n} \\ Y_{31} & Y_{32} & Y_{33} & \text{.....} & Y_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ Y_{n1} & Y_{n2} & Y_{n3} & \text{.....} & Y_{nn} \end{bmatrix} \quad \text{.....(1.53)}$$

The bus admittance matrix, Y_{bus} is symmetrical around the principal diagonal. The admittances $Y_{11}, Y_{22}, Y_{33}, \ldots, Y_{nn}$ are called self admittances at the buses and the other admittances are called mutual admittances.

In general,

Y_j = Sum of all admittances connected to bus-j.

Y_{jk} = Negative of sum of all admittances connected between bus-j and bus-k.

Also, $Y_{jk} = Y_{kj}$

Note : If the power system is represented by reactance diagram, then all the elements of the network are inductive susceptances (which are negative). In this case, Y_j will be negative and Y_{jk} will be positive.

Solution of bus voltages

Consider the node basis matrix equation [equ(1.52)] of N-bus system.

$$Y_{bus}V = I \quad \text{.....(1.54)}$$

On premultiplying the equ(1.54) by Y_{bus}^{-1} we get,

$$V = Y_{bus}^{-1}I \quad \text{.....(1.55)}$$

We know that, $Y_{bus}^{-1} = \dfrac{\text{Adjoint of } Y_{bus}}{\text{Determinant of } Y_{bus}}$

Let Δ = Determinant of Y_{bus}
and Δ_{jk} = Cofactor of Y_{jk}

$$\text{Now, Adjoint of } Y_{bus} = \begin{bmatrix} \Delta_{11} & \Delta_{12} & \Delta_{13} \text{ ..... } \Delta_{1n} \\ \Delta_{21} & \Delta_{22} & \Delta_{23} \text{ ..... } \Delta_{2n} \\ \Delta_{31} & \Delta_{32} & \Delta_{33} \text{ ..... } \Delta_{3n} \\ \vdots & \vdots & \vdots \quad \vdots \\ \Delta_{n1} & \Delta_{n2} & \Delta_{n3} \text{ ..... } \Delta_{nn} \end{bmatrix}^T = \begin{bmatrix} \Delta_{11} & \Delta_{21} & \Delta_{31} \text{ ..... } \Delta_{n1} \\ \Delta_{12} & \Delta_{22} & \Delta_{32} \text{ ..... } \Delta_{n2} \\ \Delta_{13} & \Delta_{23} & \Delta_{33} \text{ ..... } \Delta_{n3} \\ \vdots & \vdots & \vdots \quad \vdots \\ \Delta_{1n} & \Delta_{2n} & \Delta_{3n} \text{ ..... } \Delta_{nn} \end{bmatrix} \quad \text{—(1.56)}$$

$$\text{Also, } V = \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ \vdots \\ V_n \end{bmatrix} \quad \text{and} \quad I = \begin{bmatrix} I_{11} \\ I_{12} \\ I_{13} \\ \vdots \\ I_{nn} \end{bmatrix} \qquad \text{.....(1.57)}$$

Using equations (1.56) and (1.57), the equation (1.55) can be written as shown in equ(1.58).

$$\begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ \vdots \\ V_n \end{bmatrix} = \frac{1}{\Delta} \begin{bmatrix} \Delta_{11} & \Delta_{21} & \Delta_{31} & \ldots\ldots & \Delta_{n1} \\ \Delta_{12} & \Delta_{22} & \Delta_{32} & \ldots\ldots & \Delta_{n2} \\ \Delta_{13} & \Delta_{23} & \Delta_{33} & \ldots\ldots & \Delta_{n3} \\ \vdots & \vdots & \vdots & \vdots & \\ \Delta_{1n} & \Delta_{2n} & \Delta_{3n} & \ldots\ldots & \Delta_{nn} \end{bmatrix} \begin{bmatrix} I_{11} \\ I_{22} \\ I_{33} \\ \vdots \\ I_{nn} \end{bmatrix} \qquad \text{.....(1.58)}$$

By matrix multiplication the equation (1.58) can be expressed as n-number of linear independent equations shown below.

$$V_1 = \frac{1}{\Delta}\left[\Delta_{11}I_{11} + \Delta_{21}I_{22} + \Delta_{31}I_{33} + \ldots\ldots + \Delta_{n1}I_{nn}\right]$$

$$V_2 = \frac{1}{\Delta}\left[\Delta_{12}I_{11} + \Delta_{22}I_{22} + \Delta_{32}I_{33} + \ldots\ldots + \Delta_{n2}I_{nn}\right]$$

$$V_3 = \frac{1}{\Delta}\left[\Delta_{13}I_{11} + \Delta_{23}I_{22} + \Delta_{33}I_{33} + \ldots\ldots + \Delta_{n3}I_{nn}\right]$$

$$\vdots \qquad\qquad \vdots$$

$$V_n = \frac{1}{\Delta}\left[\Delta_{1n}I_{11} + \Delta_{2n}I_{22} + \Delta_{3n}I_{33} + \ldots\ldots + \Delta_{nn}I_{nn}\right]$$

In general the kth bus voltage is given by,

$$V_k = \frac{1}{\Delta}\left[\Delta_{1k}I_{11} + \Delta_{2k}I_{22} + \Delta_{3k}I_{33} + \ldots\ldots + \Delta_{nk}I_{nn}\right]$$

$$\therefore V_k = \frac{1}{\Delta}\sum_{j=1}^{n} \Delta_{jk}\, I_{jj} \qquad \text{.....(1.59)}$$

where, Δ = Determinant of Y_{bus} matrix.

I_{jj} = Sum of current sources injecting current to node-j.

Δ_{jk} = Cofactor of the element Y_{jk} of bus admittance matrix.

Note : *The equ(1.59) is Cramer's rule and this equation can be expressed in another simpler form as shown below.*

Let Δ = Determinant of Y_{bus} matrix.

Δ_k = Determinant of Y_{bus} matrix after replacing k^{th} column by current source vector I.

Now, k^{th} bus voltage, $V_k = \frac{\Delta_k}{\Delta}$

Bus or node elimination by matrix algebra

The buses or nodes which does not have any current sources can be eliminated by matrix manipulation of the standard node equation.

Consider the general form of node basis matrix equation,

$$Y_{bus}V = I \qquad \text{.....(1.60)}$$

Now the matrices in the equ(1.60) can be partitioned using the guidelines given below.

1. *The column matrices V and I are rearranged such that the element associated with buses to be eliminated are in the lower rows of the matrices.*
2. *The square matrix Y_{bus} is rearranged such that the elements associated with buses to be eliminated are in the last rows and columns of the matrices.*
3. *The bus admittance matrix is partitioned so that elements identified only with nodes to be eliminated are separated from the other elements by horizontal and vertical lines.*

Consider a bus admittance matrix of order (5×5). Now the matrix equ(1.60) for n = 5 can be written as shown in equ(1.61). Let us assume that the buses 4 and 5 does not have any current source and so they can be eliminated.

$$\left[\begin{array}{ccc:cc} Y_{11} & Y_{12} & Y_{13} & Y_{14} & Y_{15} \\ Y_{21} & Y_{22} & Y_{23} & Y_{24} & Y_{25} \\ Y_{31} & Y_{32} & Y_{33} & Y_{34} & Y_{35} \\ \hdashline Y_{41} & Y_{42} & Y_{43} & Y_{44} & Y_{45} \\ Y_{51} & Y_{52} & Y_{53} & Y_{54} & Y_{55} \end{array}\right] \left[\begin{array}{c} V_1 \\ V_2 \\ V_3 \\ \hdashline V_4 \\ V_5 \end{array}\right] = \left[\begin{array}{c} I_{11} \\ I_{22} \\ I_{33} \\ \hdashline I_{44} \\ I_{55} \end{array}\right] \qquad \text{.....(1.61)}$$

Let us partition the matrix equ(1.61) and define the following submatrices.

$$K = \begin{bmatrix} Y_{11} & Y_{12} & Y_{13} \\ Y_{21} & Y_{22} & Y_{23} \\ Y_{31} & Y_{32} & Y_{33} \end{bmatrix} \quad L = \begin{bmatrix} Y_{14} & Y_{15} \\ Y_{24} & Y_{25} \\ Y_{34} & Y_{35} \end{bmatrix} \quad L^T = \begin{bmatrix} Y_{41} & Y_{42} & Y_{43} \\ Y_{51} & Y_{52} & Y_{53} \end{bmatrix}$$

$$M=\begin{bmatrix}Y_{44} & Y_{45}\\ Y_{54} & Y_{55}\end{bmatrix} \qquad V_A=\begin{bmatrix}V_1\\ V_2\\ V_3\end{bmatrix} \qquad V_X=\begin{bmatrix}V_4\\ V_5\end{bmatrix} \qquad I_A=\begin{bmatrix}I_{11}\\ I_{22}\\ I_{33}\end{bmatrix} \qquad I_X=\begin{bmatrix}I_{44}\\ I_{55}\end{bmatrix}$$

where, I_X = Submatrix composed of the currents entering the buses to be eliminated.

V_X = Submatrix composed of the voltages of the buses to be eliminated.

K = Submatrix composed of self and mutual admittances identified only with buses to be retained.

M = Submatrix composed of self and mutual admittances identified only with buses to be eliminated.

L = Submatrix composed of only mutual admittances between buses to be retained and eliminated.

L^T = Transpose of L.

Using the submatrices as defined above, the matrix equation (1.61) can be written as shown in equ (1.62).

$$\begin{bmatrix}K & L\\ L^T & M\end{bmatrix}\begin{bmatrix}V_A\\ V_X\end{bmatrix}=\begin{bmatrix}I_A\\ I_X\end{bmatrix} \qquad \text{.....(1.62)}$$

By matrix multiplication the equ(1.62) can be written as,

$$K V_A + L V_X = I_A$$

$$L^T V_A + M V_X = I_X \qquad \text{.....(1.63)}$$

From equ(1.64) we get, ...(1.64)

$$M V_X = I_X - L^T V_A \qquad \text{.....(1.65)}$$

Here all the elements of the submatrix I_X are zero, because the buses to be eliminated does not have any current source.

$$\therefore M V_X = -L^T V_A$$

On premultiplying equ(1.66) by M^{-1} we get, ...(1.66)

$$V_X = -M^{-1} L^T V_A$$

On substituting for V_X from equ (1.67) in equ(1.63) we get, ...(1.67)

$$K V_A - L M^{-1} L^T V_A = I_A$$

$$\therefore [K - L M^{-1} L^T] V_A = I_A$$

$$Y_{bus,new} V_A = I_A$$

where, $Y_{bus,new} = K - L M^{-1} L^T$ .....(1.68)

.....(1.69)

The new bus admittance matrix $Y_{bus,new}$ used to reconstruct the circuit with the unwanted buses eliminated.

The matrix partitioning method discussed above is a general procedure and it is suitable for computer solutions. When large number of buses are to be eliminated, then the size of matrix M will be large and finding M^{-1} will be tedious.

The process of bus elimination can be simplified if one bus is eliminated at a time. In this case the matrix M will have only one element and M^{-1} is the reciprocal of this element. Here, the bus to be eliminated must be the highest numbered bus and renumbering may be required. This can be achieved by row interchange and column interchange. When bus-p has to be eliminated in a system with n-independent buses, the pth row is interchanged with n^{th} row and p^{th} column is interchanged with n^{th} column in the bus admittance matrix..

Consider a bus admittance matrix of order (n×n), in which the nth bus has to be eliminated

$$Y_{bus}=\begin{bmatrix}Y_{11} & Y_{12} & \cdots & Y_{1(n-1)} & Y_{1n}\\ Y_{21} & Y_{22} & \cdots & Y_{2(n-1)} & Y_{2n}\\ \vdots & \vdots & & \vdots & \vdots\\ Y_{(n-1)1} & Y_{(n-2)2} & \cdots & Y_{(n-1)(n-1)} & Y_{(n-1)n}\\ Y_{n1} & Y_{n2} & \cdots & Y_{n(n-1)} & Y_{nn}\end{bmatrix} \qquad \text{.....(1.70)}$$

(K: first $n-1$ rows, first $n-1$ columns; L: first $n-1$ rows, last column; L^T: last row, first $n-1$ columns; M: last row, last column)

Let us partition the bus admittance matrix as shown in equ(1.70). The submatrices are given below.

$$K=\begin{bmatrix}Y_{11} & Y_{12} & \cdots & Y_{1(n-1)}\\ Y_{21} & Y_{22} & \cdots & Y_{2(n-1)}\\ \vdots & \vdots & & \vdots\\ Y_{(n-1)1} & Y_{(n-2)2} & \cdots & Y_{(n-1)(n-1)}\end{bmatrix} \quad ; \quad L=\begin{bmatrix}Y_{1n}\\ Y_{2n}\\ \vdots\\ Y(n-1)n\end{bmatrix}$$

$$L^T=\begin{bmatrix}Y_{n1} & Y_{n2} & \cdots & Y_{n(n-1)}\end{bmatrix} \quad ; \quad M=Y_{nn} \quad ; \quad M^{-1}=\frac{1}{Y_{nn}}$$

From equ(1.69) we get,

$$Y_{bus,new} = K - L M^{-1} L^T \qquad \text{.....(1.71)}$$

On substituting the submatrices in equ(1.71) we get,

$$Y_{bus,new} = \begin{bmatrix} Y_{11} & Y_{12} & \ldots\ldots & Y_{1(n-1)} \\ Y_{21} & Y_{22} & \ldots\ldots & Y_{2(n-1)} \\ \vdots & \vdots & & \vdots \\ Y_{(n-1)1} & Y_{(n-2)2} & \ldots\ldots & Y_{(n-1)(n-1)} \end{bmatrix} - \begin{bmatrix} Y_{1n} \\ Y_{2n} \\ \vdots \\ Y(n-1)n \end{bmatrix} \left[\frac{1}{Y_{nn}}\right] \begin{bmatrix} Y_{n1} & Y_{n2} & \ldots\ldots & Y_{n(n-1)} \end{bmatrix}$$

$$Y_{bus,new} = \begin{bmatrix} Y_{11} & Y_{12} & \ldots\ldots & Y_{1(n-1)} \\ Y_{21} & Y_{22} & \ldots\ldots & Y_{2(n-1)} \\ \vdots & \vdots & & \vdots \\ Y_{(n-1)1} & Y_{(n-2)2} & \ldots\ldots & Y_{(n-1)(n-1)} \end{bmatrix} - \frac{1}{Y_{nn}} \begin{bmatrix} Y_{1n}Y_{n1} & Y_{1n}Y_{n2} & \ldots\ldots & Y_{1n}Y_{n(n-1)} \\ Y_{2n}Y_{n1} & Y_{2n}Y_{n2} & \ldots\ldots & Y_{2n}Y_{n(n-1)} \\ \vdots & \vdots & & \vdots \\ Y_{(n-1)n}Y_{n1} & Y_{(n-1)n}Y_{n2} & \ldots\ldots & Y_{(n-1)n}Y_{n(n-1)} \end{bmatrix} \quad \ldots\ldots(1.72)$$

From equation (1.72) the element Y_{jk} (i.e. the element in row-k and column-j) of the resulting $(n-1) \times (n-1)$ new bus admittance matrix is given by

$$Y_{jk,new} = Y_{jk} - \frac{Y_{jn} Y_{nk}}{Y_{nn}} \quad \ldots\ldots(1.73)$$

for $j = 1, 2, 3, \ldots, (n-1)$ and $k = 1, 2, 3, \ldots, (n-1)$.

where Y_{jk}, Y_{jn}, Y_{nk} and Y_{nn} are elements of original or given bus admittance matrix.

The equ(1.73) can be used to compute the elements of new bus admittance matrix directly from the elements of original bus admittance matrix.

The following procedure can be used to compute an element of bus admittance matrix using equ(1.73).

Step 1 : *Consider an element Y_{jk} of original bus admittance matrix.*

Step 2 : *Get the product of last element of row-j and last element of column-k.*

Step 3 : *Divide the product obtained in step-2 by Y_{nn} of original bus admittance matrix.*

Step 4 : *On substracting the resultant value obtained in step-3 from Y_{jk} of original bus admittance matrix we get, the new value of Y_{jk}.*

Note : Since bus admittance matrix is symmetrical we can take $Y_{kj,new} = Y_{jk,new}$ and can avoid m(m-1)/2 calculations for a new bus admittance matrix of order (mxm), where m = n-1.

EXAMPLE 1.13

Solve the node voltages V_1 and V_2 in the network shown in fig 1.13.1. The voltages and impedances are in p.u.

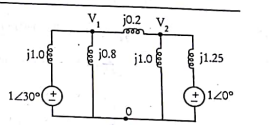


Fig 1.13.1.

SOLUTION

The voltage sources in the network can be converted to current sources by source transformation as shown in fig 1.13.2.

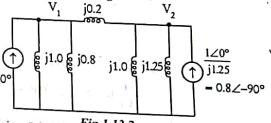


Fig 1.13.2.

The node basis matrix equation is formed as shown below.

$$\begin{bmatrix} \frac{1}{j1}+\frac{1}{j0.8}+\frac{1}{j0.2} & -\frac{1}{j0.2} \\ -\frac{1}{j0.2} & \frac{1}{j0.2}+\frac{1}{j1}+\frac{1}{j1.25} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} \frac{1\angle 30°}{j1} \\ \frac{1\angle 0°}{j1.25} \end{bmatrix}$$

$$\begin{bmatrix} -j1 - j1.25 - j5 & j5 \\ j5 & -j5 - j - j0.8 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 1\angle 30° \\ 1\angle 90° \\ \hline 1\angle 0° \\ 1.25\angle 90° \end{bmatrix}$$

$$\begin{bmatrix} -j7.25 & j5 \\ j5 & -j6.8 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 1\angle -60° \\ 0.8\angle -90° \end{bmatrix}$$

$$\Delta = \begin{vmatrix} -j7.25 & j5 \\ j5 & -j6.8 \end{vmatrix} = (-j7.25)(-j6.8) - (j5)(j5)$$

$$= j^2(7.25 \times 6.8) - j^2 25 = -(7.25 \times 6.8) + 25 = -24.3$$

In node basis analysis; k^{th} node voltage V_k, is given by,

$$V_k = \frac{1}{\Delta} \sum_{j=1}^{n} \Delta_{jk} I_{jj}$$

where, Δ_{jk} = Cofactor of Y_{jk}

Δ = Determinant of Y

I_{jj} = Sum of current sources connected to node j.

Here n = 2

$$\therefore V_1 = \frac{1}{\Delta}\left[\Delta_{11} I_{11} + \Delta_{21} I_{22}\right]$$

$$= \frac{-1}{24.3}\left[(-j0.8 \times 1\angle -60°) + (-j5 \times 0.8\angle -90°)\right]$$

$$= -\frac{1}{24.3}\left[(6.8\angle -90° \times 1\angle -60°) + (5\angle -90° \times 0.8\angle -90°)\right]$$

$$= -\frac{1}{24.3}\left[(6.8\angle -150° + 4\angle -180°\right] = -\frac{1}{24.3}\left[-5.88 - j3.4 - 4\right]$$

$$= \frac{-9.88 - j3.4}{-24.3} = \frac{9.88}{24.3} + j\frac{3.4}{24.3} = 0.4066 + j0.1399 = 0.43\angle 19° \text{ p.u.}$$

$$V_2 = \frac{1}{\Delta}\left[\Delta_{12} I_{11} + \Delta_{22} I_{22}\right]$$

$$= \frac{-1}{24.3}\left[(-j5 \times 1\angle -60°) + (-j7.25 \times 0.8\angle -90°)\right]$$

$$= -\frac{1}{24.3}\left[(5\angle -90° \times 1\angle -60°) + (7.25\angle -90° \times 0.8\angle -90°)\right]$$

$$= \frac{5\angle -150° + 5.8\angle -180°}{-24.3} = \frac{-4.33 - j2.5 - 5.8}{-24.3}$$

$$= \frac{-10.13 - j2.5}{-24.3} = \frac{10.13}{24.3} + j\frac{2.5}{24.3} = 0.4169 + j0.1029 = 0.43\angle 14° \text{ p.u.}$$

EXAMPLE 1.14

For the network shown in fig 1.14.1. form the bus admittance matrix. Determine the reduced admittance matrix by eliminating node 4. The values are marked in p.u.

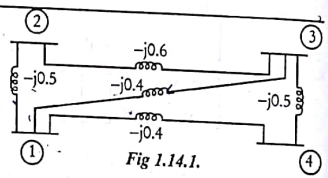


Fig 1.14.1.

SOLUTION

The Y_{bus} matrix of the network is,

$$Y_{bus} = \begin{bmatrix} -(j0.5 + j0.4 + j0.4) & j0.5 & j0.4 & j0.4 \\ j0.5 & -(j0.5 + j0.6) & j0.6 & 0 \\ j0.4 & j0.6 & -(j0.6 + j0.5 + j0.4) & j0.5 \\ j0.4 & 0 & j0.5 & -(j0.5 + j0.4) \end{bmatrix}$$

$$Y_{bus} = \begin{bmatrix} -j1.3 & j0.5 & j0.4 & j0.4 \\ j0.5 & -j1.1 & j0.6 & 0 \\ j0.4 & j0.6 & -j1.5 & j0.5 \\ j0.4 & 0 & j0.5 & -j0.9 \end{bmatrix}$$

The elements of new bus admittance matrix after eliminating the 4th row and 4th column is given by

$$Y_{jk,\,new} = Y_{jk} - \frac{Y_{jn}\, Y_{nk}}{Y_{nn}} \;;\; \text{where } n = 4 \;;\; j = 1, 2, 3 \text{ and } k = 1, 2, 3,$$

The bus admittance matrix is symmetrical, $\therefore Y_{kj,\,new} = Y_{jk,\,new}$

$$Y_{11,\,new} = Y_{11} - \frac{Y_{14}\, Y_{41}}{Y_{44}} = -j1.3 - \frac{(j0.4)\,(j0.4)}{-j0.9} = -j1.12$$

$$Y_{12,\,new} = Y_{12} - \frac{Y_{14}\, Y_{42}}{Y_{44}} = j0.5 - \frac{(j0.4 \times 0)}{-j0.9} = j0.5$$

$$Y_{13,\,new} = Y_{13} - \frac{Y_{14}\, Y_{43}}{Y_{44}} = j0.4 - \frac{(j0.4)\,(j0.5)}{-j0.9} = j0.622$$

$$Y_{21,\,new} = Y_{12,\,new} = j0.5$$

$$Y_{22,\,new} = Y_{22} - \frac{Y_{24}\, Y_{42}}{Y_{44}} = -j1.1 - \frac{(0)\,(0)}{-j0.9} = -j1.1$$

$$Y_{23,\,new} = Y_{23} - \frac{Y_{24}\, Y_{43}}{Y_{44}} = j0.6 - \frac{(0)\,(j0.5)}{-j0.9} = j0.6$$

$$Y_{31,\,new} = Y_{13,\,new} = j0.622$$

$$Y_{32,\,new} = Y_{23,\,new} = j0.6$$

$$Y_{33,\,new} = Y_{33} - \frac{Y_{34}\, Y_{43}}{Y_{44}} = -j1.5 - \frac{(j0.5)\,(j0.5)}{-j0.9} = -j1.222$$

The reduced bus admittance matrix after eliminating 4th row is shown below.

$$Y_{bus} = \begin{bmatrix} -j1.12 & j0.5 & j0.622 \\ j0.5 & -j1.1 & j0.6 \\ j0.622 & j0.6 & -j1.222 \end{bmatrix}$$

EXAMPLE 1.15

Eliminate buses 3 and 4 in the given bus admittance matrix and form new bus admittance matrix.

$$Y_{bus} = \begin{bmatrix} -j9.8 & 0.0 & j4.0 & j5.0 \\ 0.0 & -j8.3 & j2.5 & j5.0 \\ j4.0 & j2.5 & -j14 & j8.0 \\ j5.0 & j5.0 & j8.0 & -j18.0 \end{bmatrix}$$

SOLUTION

First let us eliminate 4th bus, $\therefore Y_{nn} = Y_{44} = -j18.0$

The elements of new bus admittance after eliminating 4th row and 4th column is given by

$$Y_{jk,\,new} = Y_{jk} - \frac{Y_{jn}\,Y_{nk}}{Y_{nn}} \quad \text{where } n = 4\ ;\ j = 1,\ 2,\ 3,\ \text{and } k = 1,\ 2,\ 3$$

$$Y_{11,\,new} = Y_{11} - \frac{Y_{14}\,Y_{41}}{Y_{44}} = -j9.8 - \frac{j5.0 \times j5.0}{-j18.0} = -j8.411$$

$$Y_{12,\,new} = Y_{12} - \frac{Y_{14}\,Y_{42}}{Y_{44}} = 0.0 - \frac{j5.0 \times j5.0}{-j18.0} = j1.388$$

$$Y_{13,\,new} = Y_{13} - \frac{Y_{14}\,Y_{43}}{Y_{44}} = j4.0 - \frac{j5 \times j8}{-j18.0} = j6.222$$

$$Y_{21,\,new} = Y_{12,\,new} = j1.388$$

$$Y_{22,\,new} = Y_{22} - \frac{Y_{24}\,Y_{42}}{Y_{44}} = -j8.3 - \frac{j5 \times j5}{-j18.0} = -j6.911$$

$$Y_{23,\,new} = Y_{23} - \frac{Y_{24}\,Y_{43}}{Y_{44}} = j2.5 - \frac{j5 \times j8}{-j18} = j4.722$$

$$Y_{31,\,new} = Y_{13,\,new} = j6.222$$

$$Y_{32,\,new} = Y_{23,\,new} = j4.722$$

$$Y_{33,\,new} = Y_{33} - \frac{Y_{34}\,Y_{43}}{Y_{44}} = -j14 - \frac{j8 \times j8}{-j18.0} = -j10.444$$

The reduced bus admittance matrix after eliminating 4th node is given by

$$Y_{bus} = \begin{bmatrix} -j8.411 & j1.3889 & j6.222 \\ j1.388 & -j6.911 & j4.722 \\ j6.222 & j4.722 & -j10.444 \end{bmatrix}$$

Elimination of node 3 : $Y_{nn} = Y_{33} = -j10.444$

The other elements of the reduced bus admittance matrix can be formed from the equation

$$Y_{jk,\,new} = Y_{jk} - \frac{Y_{jn}\,Y_{nk}}{Y_{nn}} \quad \text{where } n = 3\ ;\ j = 1,\ 2 \text{ and } k = 1,\ 2$$

$$Y_{11,\,new} = Y_{11} - \frac{Y_{13}\,Y_{31}}{Y_{33}} = -j8.411 - \frac{j6.222 \times j6.222}{-j10.444} = -j4.7043$$

$$Y_{12,\,new} = Y_{12} - \frac{Y_{13}\,Y_{32}}{Y_{33}} = j1.388 - \frac{j6.222 \times j4.722}{-j10.444} = j4.2011$$

$$Y_{21,\,new} = Y_{12,\,new} = j4.2011$$

$$Y_{22,\,new} = Y_{22} - \frac{Y_{23}\,Y_{32}}{Y_{33}} = -j6.911 - \frac{j4.722 \times j4.722}{-j10.444} = -j4.7761$$

The reduced bus admittance matrix after eliminating node 3 and 4 is

$$Y_{bus} = \begin{bmatrix} -j4.7043 & j4.2011 \\ j4.2011 & -j4.7761 \end{bmatrix}$$

EXAMPLE 1.16

Determine the bus admittance matrix of the system whose reactance diagram is shown in fig 1.16.1. The currents and admittances are given in p.u. Determine the reduced bus admittance matrix after eliminating node-3.

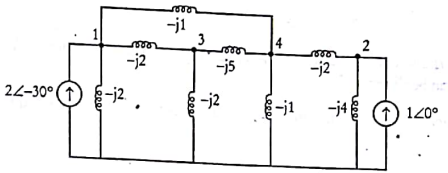


Fig 1.16.1.

SOLUTION

The bus admittance matrix can be formed by inspection using the following guidelines.

1. The diagonal element Y_{jj} is given by sum of all the admittances connected to node-j.
2. The off-diagonal element Y_{jk} is given by negative of the sum of all the admittances connected between node-j and node-k.

$$\therefore Y_{bus} = \begin{bmatrix} -j2 - j2 - j1 & 0 & j2 & j1 \\ 0 & -j2 - j4 & 0 & j2 \\ j2 & 0 & -j2 - j2 - j5 & j5 \\ j1 & j2 & j5 & -j1 - j5 - j2 - j1 \end{bmatrix}$$

$$\therefore Y_{bus} = \begin{bmatrix} -j5 & 0 & j2 & j1 \\ 0 & -j6 & 0 & j2 \\ j2 & 0 & -j9 & j5 \\ j1 & j2 & j5 & -j9 \end{bmatrix} \quad \text{.....(1.16.1)}$$

For eliminating node-3, the bus admittance matrix is rearranged by interchanging row-3 & row-4, and then interchanging column-3 & column-4.

After interchanging row - 3 & row - 4 of Y_{bus} matrix of equ(1.16.1) we get,

$$Y_{bus} = \begin{bmatrix} -j5 & 0 & j2 & j1 \\ 0 & -j6 & 0 & j2 \\ j1 & j2 & j5 & -j9 \\ j2 & 0 & -j9 & j5 \end{bmatrix} \quad \text{.....(1.16.2)}$$

After interchanging column - 3 & column - 4 of Y_{bus} matrix of equ(1.16.2) we get,

$$Y_{bus} = \begin{bmatrix} -j5 & 0 & j1 & j2 \\ 0 & -j6 & j2 & 0 \\ j1 & j2 & -j9 & j5 \\ j2 & 0 & j5 & -j9 \end{bmatrix} \quad \text{.....(1.16.3)}$$

Now the last row and last column [i.e., 4th row and 4th column] of Y_{bus} matrix of equ(1.16.3) can be eliminated.

The elements of new bus admittance matrix after eliminating 4th row and 4th column is given by

$$Y_{jk,\,new} = Y_{jk} - \frac{Y_{jn}\,Y_{nk}}{Y_{nn}} \quad \text{where } n = 4 \;;\; j = 1,2,3 \text{ and } k = 1,2,3$$

$$Y_{11,\,new} = Y_{11} - \frac{Y_{14}\,Y_{41}}{Y_{44}} = -j5 - \frac{(j2)\,(j2)}{-j9} = -j4.556$$

$$Y_{12,\,new} = Y_{12} - \frac{Y_{14}\,Y_{42}}{Y_{44}} = 0 - \frac{j2 \times 0}{-j9} = 0$$

$$Y_{13,\,new} = Y_{13} - \frac{Y_{14}\,Y_{43}}{Y_{44}} = j1 - \frac{(j2)\,(j5)}{-j9} = j2.111$$

$$Y_{21,\,new} = Y_{12,\,new} = 0$$

$$Y_{22,\,new} = Y_{22} - \frac{Y_{24}\,Y_{42}}{Y_{44}} = -j6 - \frac{0 \times 0}{-j9} = -j6$$

$$Y_{23,\,new} = Y_{23} - \frac{Y_{24}\,Y_{43}}{Y_{44}} = j2 - \frac{0 \times j5}{-j9} = j2$$

$$Y_{31,\,new} = Y_{13,\,new} = j2.111$$

$$Y_{32,\,new} = Y_{23,\,new} = j2$$

$$Y_{33,\,new} = Y_{33} - \frac{Y_{34}\,Y_{43}}{Y_{44}} = -j9 - \frac{j5 \times j5}{-j9} = -j6.222$$

The reduced bus admittance matrix after eliminating bus-3 is given by,

$$Y_{bus} = \begin{bmatrix} -j4.556 & 0 & j2.111 \\ 0 & -j6 & j2 \\ j2.111 & j2 & -j6.222 \end{bmatrix}$$

1.6 BUS IMPEDANCE MATRIX

Consider the node basis matrix equation representing the power system,

$$Y_{bus}\,V = I \quad \text{.....(1.74)}$$

On premultiplying the equ(1.74) by Y_{bus}^{-1} we get,

$$V = Y_{bus}^{-1}\,I \quad \text{.....(1.75)}$$

Now the elements of Y_{bus}^{-1} will be impedances and so the matrix Y_{bus}^{-1} can be represented by an impedance matrix called bus impedance matrix, Z_{bus}.

$$Z_{bus} = Y_{bus}^{-1} \quad \text{.....(1.76)}$$

$$\text{and } V = Z_{bus}\,I \quad \text{.....(1.77)}$$

Since the bus admittance matrix is symmetrical, the bus impedance matrix is also symmetrical around the principal diagonal. In bus impedance matrix, the elements on the main diagonal are called driving point impedance of the buses or nodes and the off-diagonal elements are called the transfer impedances of the buses or nodes. The bus impedance matrix is very useful in fault analysis or calculations.

Note : *The equ(1.77) resembles a mesh basis matrix equation but it is not so, because the matrix V is a node voltage matrix and the matrix I is a current source matrix.*

The bus impedance matrix can be determined by two methods. In one method we can form the bus admittance matrix and then taking its inverse to get the bus impedance matrix. In another method the bus impedance matrix can be directly formed from the reactance diagram and this method requires the knowledge of the modifications of existing bus impedance matrix due to addition of new bus or addition of a new line (or impedance) between existing buses.

Modification of an existing bus impedance matrix

Let us denote the original Z_{bus} of a system with n-number of independent buses as Z_{orig}. When a branch of impedance Z_b is added to the system the Z_{orig} gets modified. The branch impedance Z_b can be added to the original system in the following four different ways.

Case 1 : *Adding a branch of impedance Z_b from a new bus-p to the reference bus.*

Case 2 : *Adding a branch of impedance Z_b from a new bus-p to an existing bus-q.*

Case 3 : *Adding a branch of impedance Z_b from an existing bus-q to the reference bus.*

Case 4 : *Adding a branch of impedance Z_b between two existing buses h and q.*